

Unmasking Identity Management Architecture (IMA)

Digital Identity



O'REILLY®

Phillip J. Windley

Digital Identity

O'REILLY®

Dedication

Foreword

Preface

**Who Should
Read This
Book**

Conventions Used in This Book

Constant Width

Constant Width Bold

Constant Width Italic

Comments and Questions

Safari Enabled



Acknowledgme

Chapter 1. Intro

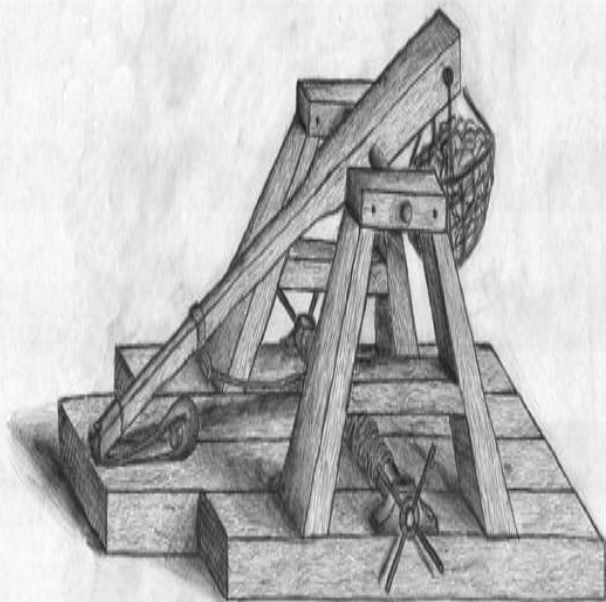


Figure 1-1. Trebuchet

Business Opportunity

Digital Identity Matters

Using Digital Identity









The Business Context of Identity

Foundational Technologies for Digital Identity

Identity Management Architectures

Chapter 2. Defir

Digital Identity

The Language

of Digital Identity

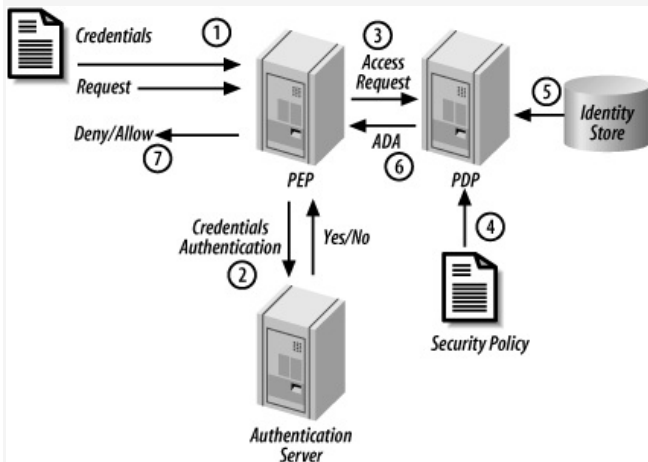


Figure 2-1. The interaction between policy enforcement points and policy decision points

Identity Scenarios in the Physical World

Identity, Security, and Privacy

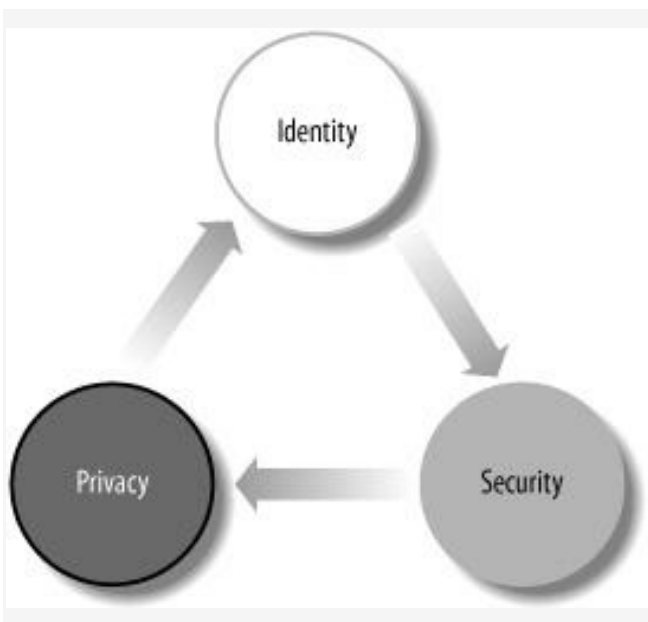


Figure 2-2. The identity, security, privacy triangle

Digital Identity Perspectives

Tier 3: Abstract Identity

Groups and demographics

Tier 2: Shared Identity

Assigned by others

Tier 1: Abstract Identity

Your permanent traits

More Detailed Information

Figure 2-3. Identity tiers and their relationship



Identity Powershifts

Conclusion

Chapter 3. Trus'





What Is Trust?







Trust and Evidence

Trust and Risk

Reputation and Trust Communities







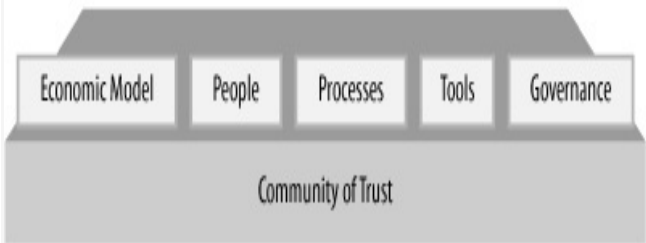


Figure 3-1. Communities of trust are supported by five components

Conclusion

Chapter 4. Privacy and Identity

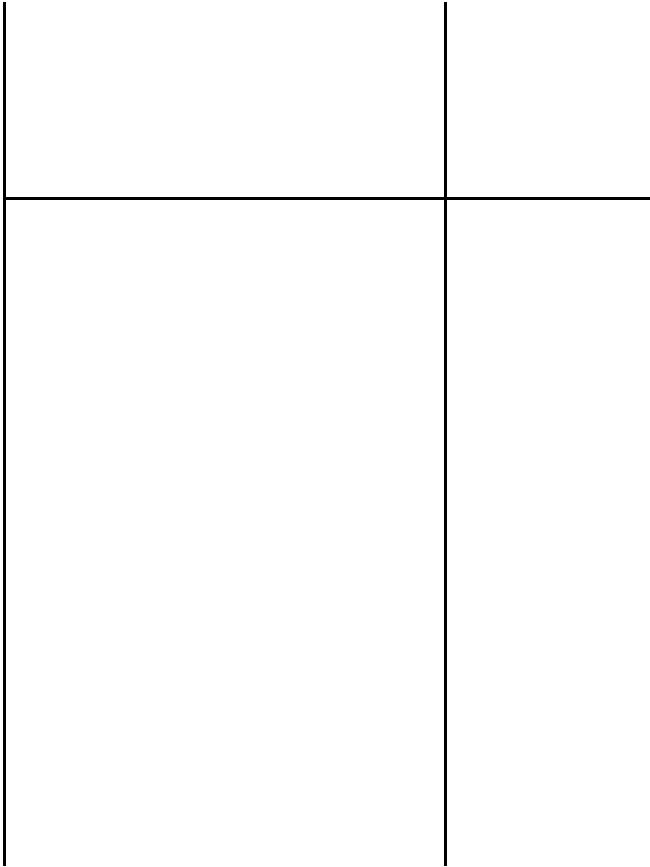
**Who's Afraid of
RFID?**

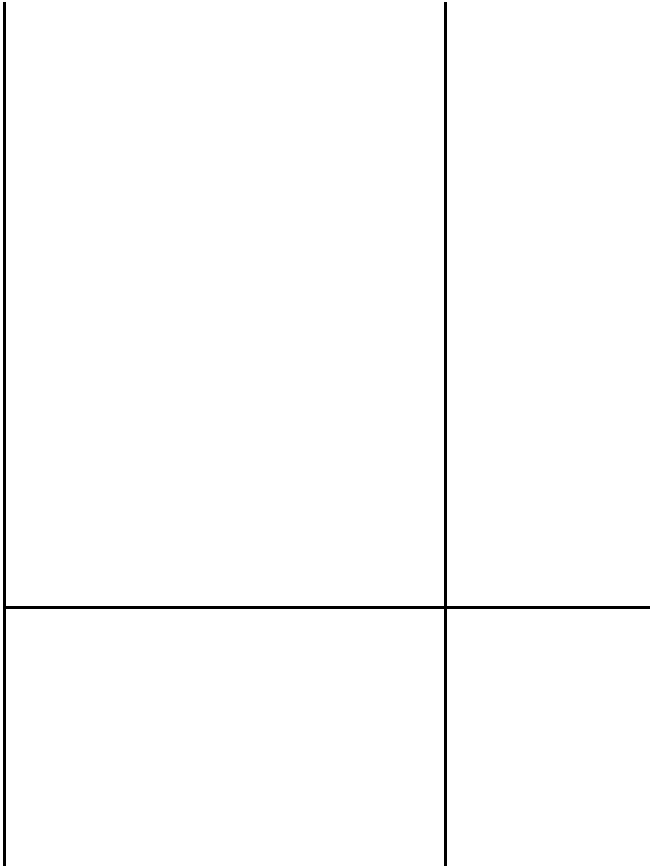
Privacy

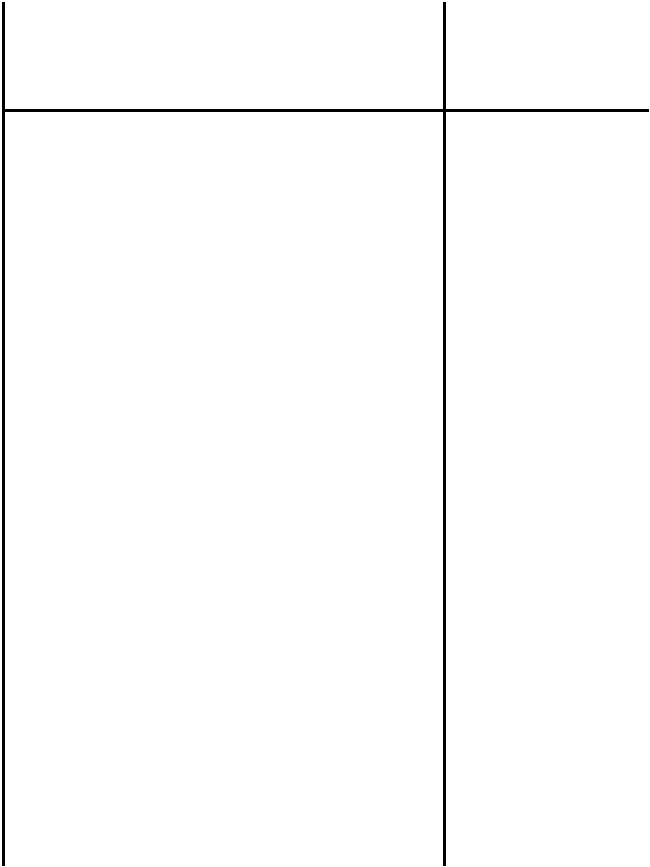
Pragmatism

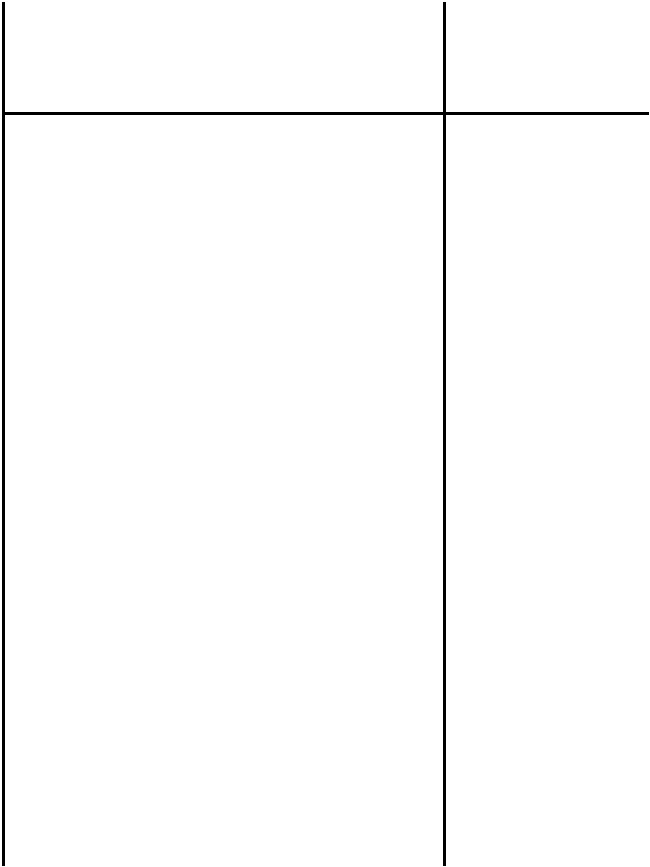
Privacy Drivers

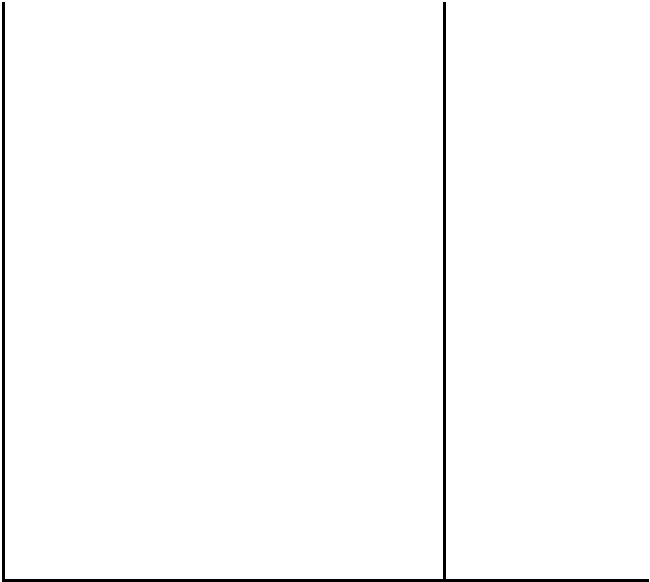
Table 4-1. Some privacy laws and regulations











Privacy Audits











Privacy Policy Capitalism









Anonymity and Pseudonymity

Privacy Principles

Prerequisites

Conclusion

Chapter 5. The Digital Identity Lifecycle

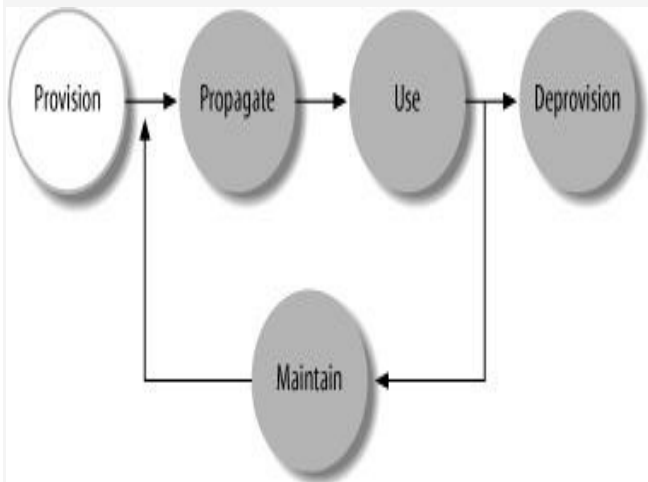


Figure 5-1. Digital identity management lifecycle

Provisioning

Propagating

Using

Maintaining

Deprovisioning

Conclusion

Chapter 6. Integrity Non- Repudiation, and Confidentiality

Integrity

Non- Repudiation

Confidentiality

Cryptography

Secret keys

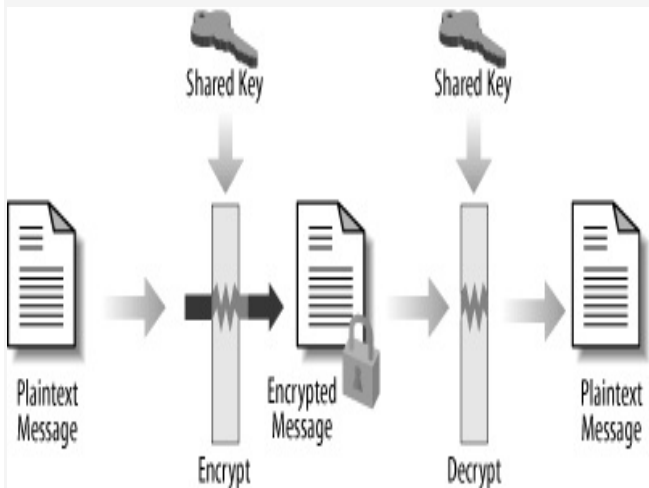


Figure 6-1. Encryption and decryption using a secret key

Table 6-1. Secret key algorithms

Public key cryptography

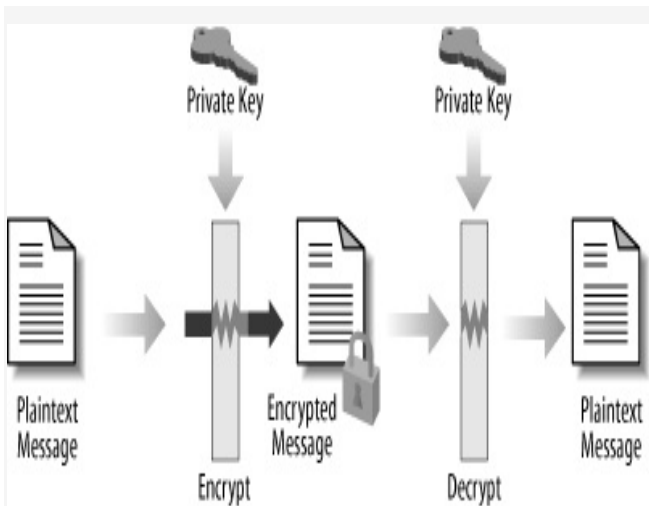


Figure 6-2. Public-key cryptography

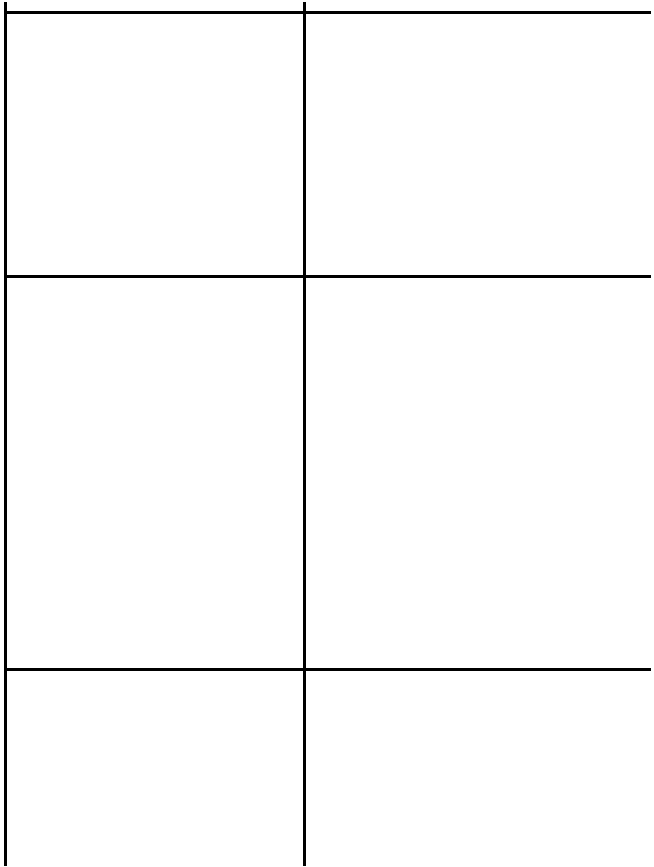


Hybrid key systems

Public key cryptosystem

algorithms

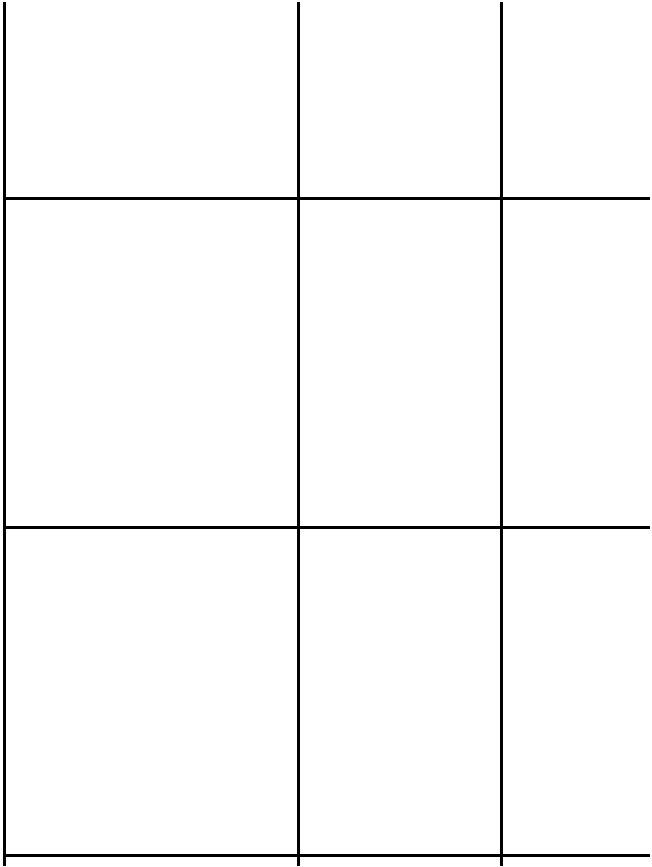
Table 6-2. Public key algorithms

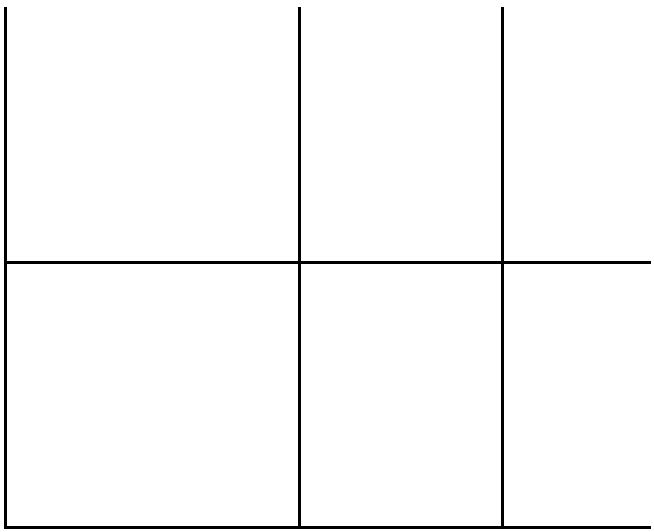




Message Digests and Hashes

Table 6-3. Message digest algorithms





Digital Signatures





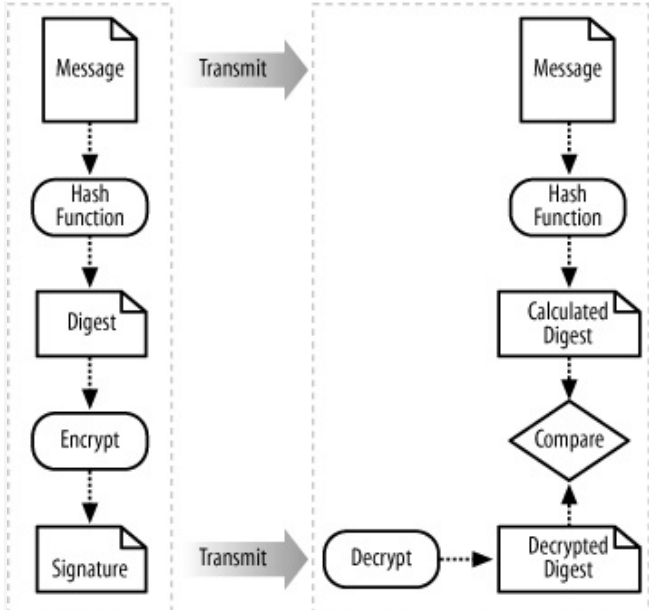


Figure 6-3. Digitally signing and verifying a message

Digital Certificates



Certificate:

Data:

Version: 3 (0x2)
Serial Number: 163

(0xa3)

Signature Algorithm:
md5WithRSAEncryption

Issuer: C=US,
ST=California, L=San Francisco,
O=Pacific

Distribution,
Inc., OU=Certification,
CN=PDI

Certificate
Authority/Email=certs@pdi.com

Validity

Not Before: Aug
19 17:31:59 2003 GMT

Not After : Aug
18 17:31:59 2004 GMT

Subject: C=US,
ST=California, L=San Francisco,

O=www.California_distribution.com,
OU=1885658,

CN=www.California_distribution.com

Subject Public Key

Info:

Public Key

Algorithm: rsaEncryption

RSA Public Key:

(512 bit)

Modulus (512

bit):

00:a4:45:15:66:ca:4b:e9:4f:b2:85:3

01:b6:a0:45:31:c4:db:67:7b:ed:29:7

8f:0e:33:3c:cb:bc:2d:2b:a8:51:48:a

82:aa:d1:07:f9:64:f2:fd:67:5d:cd:5

3d:7e:d7:ec:63

Exponent:

65537 (0x10001)

X509v3 extensions:

X509v3 Basic

Constraints:

CA:FALSE

X509v3 Key Usage:

Digital

Signature, Non Repudiation,

Key

Encipherment

X509v3 Extended

Key Usage:

TLS Web

Server Authentication

Netscape Comment:

OpenSSL

Generated Certificate.

X509v3 Subject

Key Identifier:

48:7E:92:A1:6E:B0:06:42:05:6C:FA:F

3A:2D:05:BB

X509v3 Authority

Key Identifier:

keyid:F8:F1:41:1C:D6:A2:C3:77:B1:A

E8:86:C0:0A:2B:4A

DirName:/C=US/ST=California/L=San
Francisco

/O=Pacific Distribution, Inc.

/OU=Certification/CN=PDI Cert
Authority

/Email=certs@pdi.com

serial:00

Signature Algorithm:

md5WithRSAEncryption

75:a5:ca:a3:90:e3:03:a6:40:5c:95:a
e9:fb:e8:ce:d1:b4:f5:e1:ee:04:c1:e
59:d6:d9:81:34:ec:b7:30:0a:b3:49:0
71:bd:c8:de:33:e3:f1:5f:78:67:db:4
cf:43:19:ab:4d:d4:7e:1a:2e:30:35:4
13:a9:a8:57:63:9d:0e:54:ec:16:2d:a
38:84:54:2f:7c:05:fc:9b:6b:3f:c8:8
4d:7c

Certificate Authorities

Certificate Revocations Lists







Public-Key Infrastructures

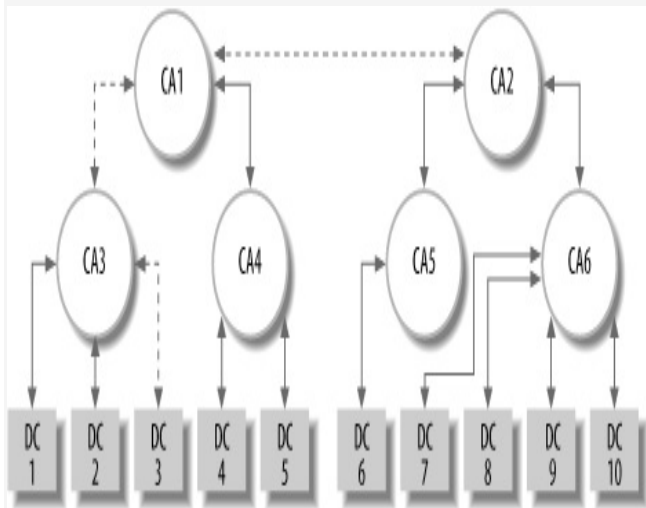


Figure 6-4. Hierarchical structures of CAs make up a PKI

Going Further

Conclusion

Chapter 7. Auth

Authentication and Trust

Authentication Systems



Cookies









ID and Password

Password management









Password reset

Challenge- Response Systems

Digital Certificates

Biometric Devices

Smart Cards

Authentication System Properties

Practicality

**Appropriate Level
of Security**

Locational Transparency

Protocol Insensitivity

Appropriate Level

of Privacy

Reliability

Auditability

Manageability

Federation Support

Conclusion

Chapter 8. Access Control





Policy First







Responsibility

Principle of Least Privilege





Accountability

Scales Better than Enforcement

Authorization Patterns

Mandatory and Discretionary Access Control

User-Based Permission Systems

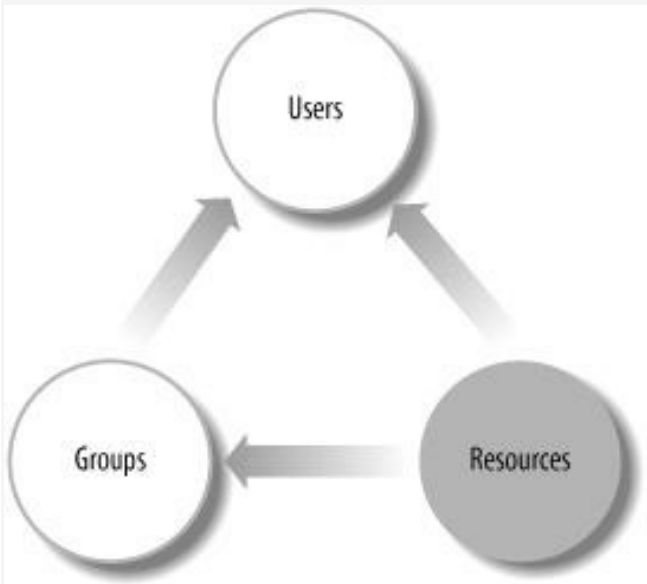


Figure 8-1. Permission hierarchy for users and groups











Access-Control Lists

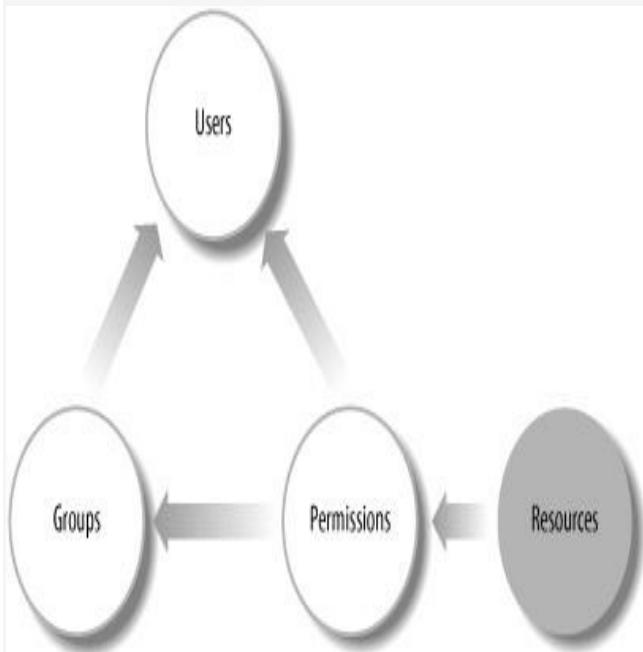


Figure 8-2. Permissions hierarchy for access-control lists

Role-Based Access Control

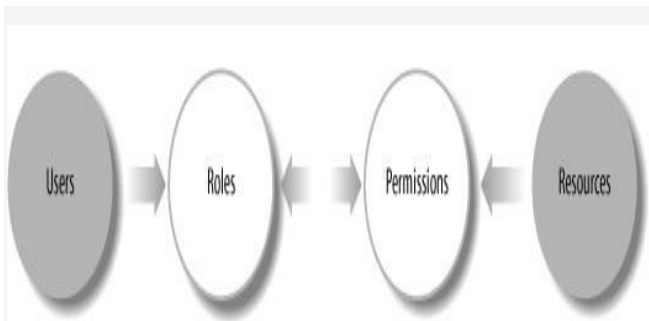


Figure 8-3. Permission

hierarchy for role-based access control





Developers

Designers

Engineering

Employee

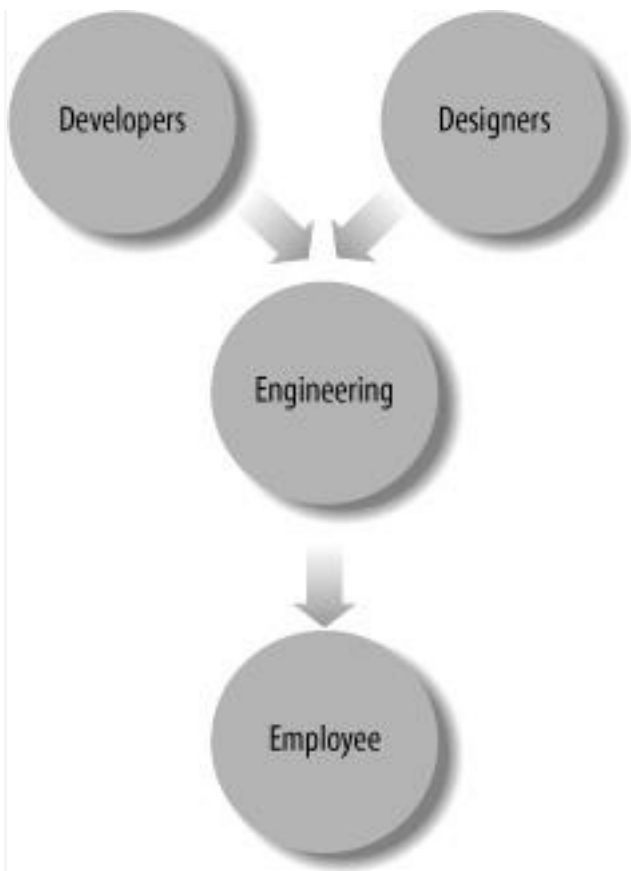


Figure 8-4. Role hierarchy for an engineering department

Abstract Authorization Architectures

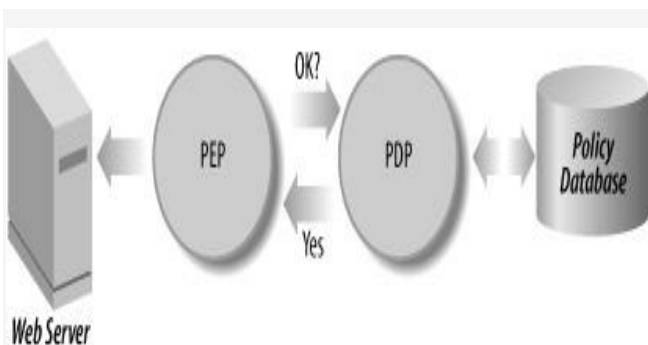


Figure 8-5. PEP and PDP actions

Digital Certificates and Access Control



Conclusion

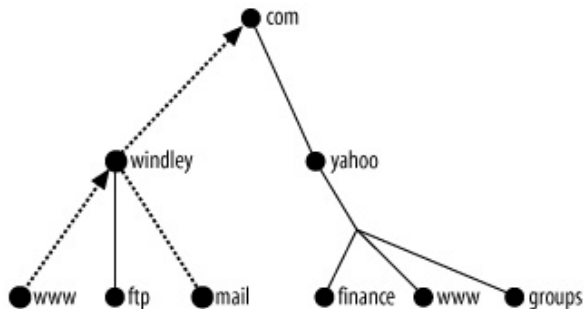
Chapter 9. Names and Directories

Utah.gov: Naming and Directories

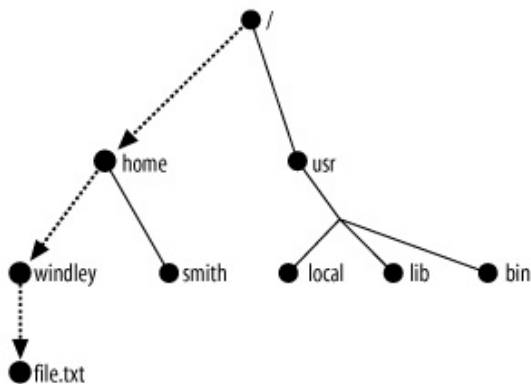


Naming

Namespaces



www.windley.com



/home/windley/file.txt

Figure 9-1. Hierarchical namespaces for domain names and filesystems









**Uniform Resource
Indicators: A
Universal**

Namespace



**Cool URIs Don't
Change**



Directories

**Directories Are Not
Databases**

















An Example Directory

Enterprise Directory Services

Domain Name System

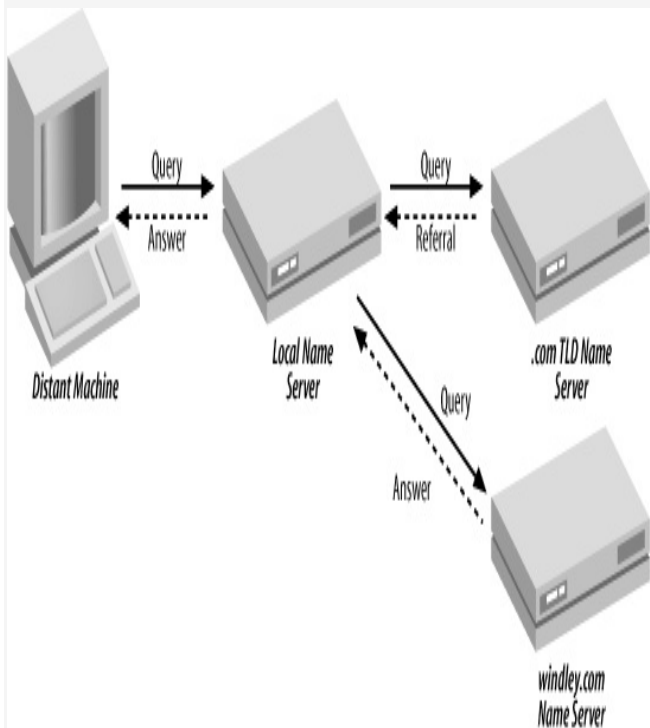


Figure 9-2. DNS query, referral, and answer

pattern for
<http://www.windley.com>

RMIRegistry





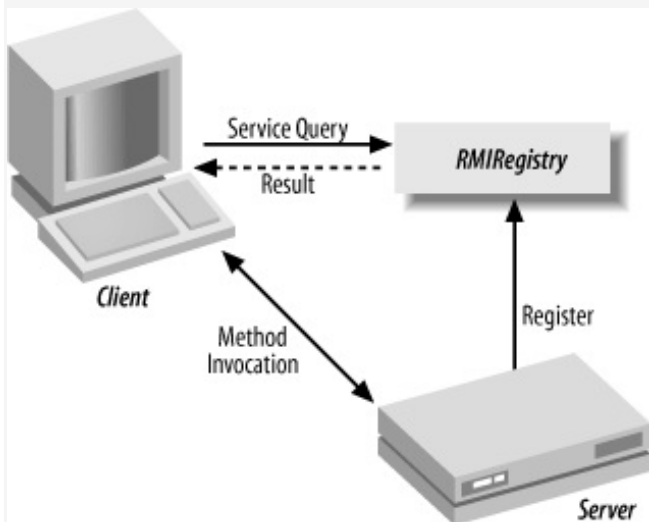


Figure 9-3. RMIRegistry is used to refer clients to services with named services

X.500: heavyweight directory services

LDAP

Aggregating Directory Information







Metadirectories





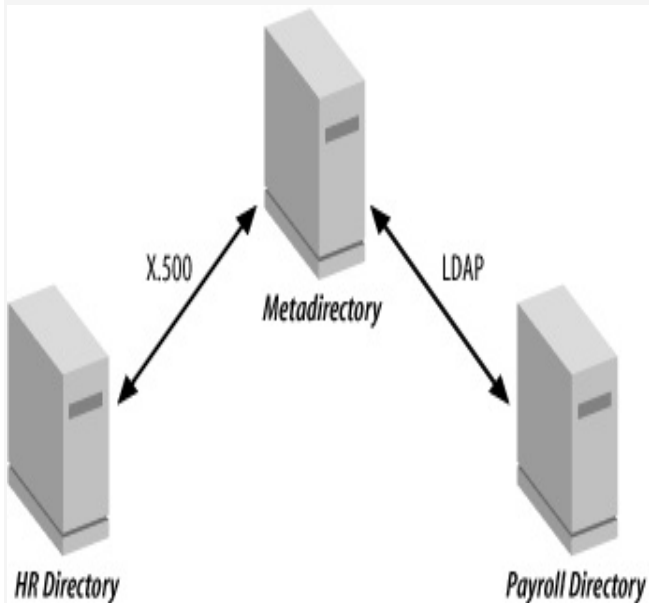


Figure 9-4. Metadirectory aggregating data from HR and Payroll directories using

different standards

Virtual Directories

Conclusion

Chapter 10. Dig Rights Management

Digital Leakage













The DRM Battle

Apple iTunes: A Case Study in DRM

Features of DRM

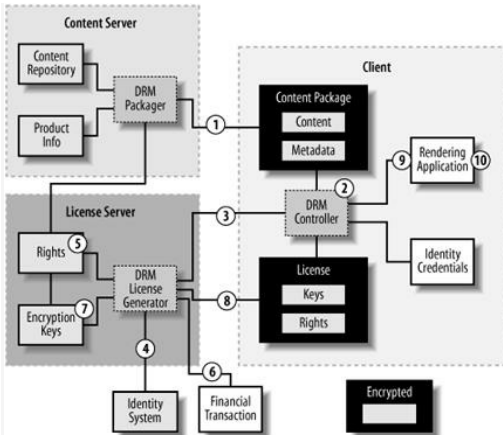








DRM Reference Architecture



Adapted from Rosenblatt et al.

Figure 10-1. A DRM reference architecture

Trusted Computing Platforms







Specifying Rights

XrML


```
<license>
  <grant>
    <keyHolder>
      <info>
        <dsig:KeyValue>
          <dsig:RSAKeyValue>
```

```
<dsig:Modulus>Fa7wo6NYf...
</dsig:Modulus>
```

```
<dsig:Exponent>AQABAA=  =
```

```
</dsig:Exponent>
    </dsig:RSAKeyValue>
  </dsig:KeyValue>
</info>
</keyHolder>
<cx:print/>
<cx:digitalWork>
  <cx:locator>
    <nonSecureIndirect
URI="http://www.contentguard.com/s

    </cx:locator>
  </cx:digitalWork>
  <validityInterval>
    <notAfter>2005-12-
24T23:59:59</notAfter>
  </validityInterval>
</grant>
</license>
```

<keyHolder/>

<print/>

<digitalWork/>

<validityInterval/>

Conclusion

Chapter 11. Inte Standards





Standards and the Digital

Identity Lifecycle

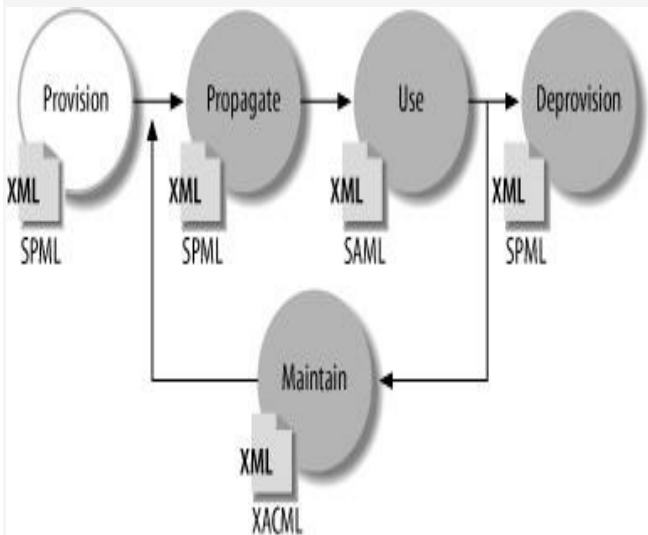


Figure 11-1. Standards and the digital identity management lifecycle

Integrity and Non- Repudiation: XML Signature

<Signature/>



<SignedInfo/>

- `<SignatureValue/>`

- `<KeyInfo/>`

<Signature
xmlns="http://www.w3.org/2000/09/x

```
<SignedInfo Id="fizz">
  <CanonicalizationMethod
    Algorithm=
      "http://www.w3.org/TR/2001/REC-
xml-c14n-20010315"/>
    <SignatureMethod
      Algorithm="http://www.w3.org/2000/
      sha1" />
      <Reference
        URI="http://www.fizzco.com/news/20
          <DigestMethod
            Algorithm="http://www.w3.org/2000/
            />
            <DigestValue>f7dhfj38sdjdf89fjskfh
            </DigestValue>
            </Reference>
```


</SignedInfo>

<SignatureValue>KLUR~HF=...Ax54</S

<KeyInfo>

<X509Data>

<X509SubjectName>

CN=Phil Windley,O=The
Windley Group LLC,ST=Utah, C=US

</X509SubjectName>

<X509Certificate>

MIID5jCCA0+gA...lVN

</X509Certificate>

</X509Data>

</KeyInfo>

</Signature>

<CanonicalizationMethod/>

<foo/>

<foo></foo>

<SignedInfo/>

<SignatureValue/>

<Reference/>

<Reference/>

<Reference/>

Confidentiality:
XML
Encryption

<EncryptedData/>

<EncryptedData/>



<EncryptionMethod/>

<KeyInfo/>

<KeyInfo/>

- <CipherData/>

<CipherValue/>

<CipherReference/>

<PaymentInfo/>

<EncryptedData/>

<PaymentInfo/>

<PaymentInfo>

 <CardName>John
Smith</CardName>
 <BillingAddress>...
</BillingAddress>
 <EncryptedData

Type='http://www.w3.org/2001/04/xml

xmlns='http://www.w3.org/2001/04/x

<CipherData>

<CipherValue>sh748fjl4...
</CipherValue>


```
    </CipherData>  
  </EncryptedData>  
</PaymentInfo>
```

```
<EncryptionMethod/>
```


Authentication and Authorization Assertions







NotBefore

NotOnOrAfter



•

•

```
<samlp:Request ...>
  <samlp:AttributeQuery>
    <saml:Subject>
```

<saml:NameIdentifier

SecurityDomain="A.com"

Name="cn=Bob"/>

</saml:Subject>

<saml:AttributeDesignator

AttributeName="Department"

AttributeNamespace="A.com"/>

</samlp:AttributeQuery>

</samlp:Request>

Bob@A.com

```
<samlp:Response
  MajorVersion="1"
MinorVersion="0"

RequestID="128.14.234.20.90123456"

InResponseTo="123.45.678.90.123456

  StatusCode="Success">

  <saml:Assertion
    MajorVersion="1"
MinorVersion="0"
```

AssertionID="123.45.678.90.1234567

Issuer="Example Company,
Inc."

IssueInstant="2003-01-
14T10:00:23Z">

<saml:Conditions
NotBefore="2003-01-
14T10:00:30Z"
NotAfter="2003-01-
14T10:15:00Z" />

<saml:AuthenticationStatement

AuthenticationMethod="Password"

AuthenticationInstant="2003-01-
14T10:00:20Z">

<saml:Subject>

<saml:NameIdentifier

```
SecurityDomain="A.com"  
    Name="cn=Alice" />  
  </saml:Subject>  
  
</saml:AuthenticationStatement>  
  </saml:Assertion>  
</samlp:Response>
```

Alice@A.com

Example SAML Use Cases

GET

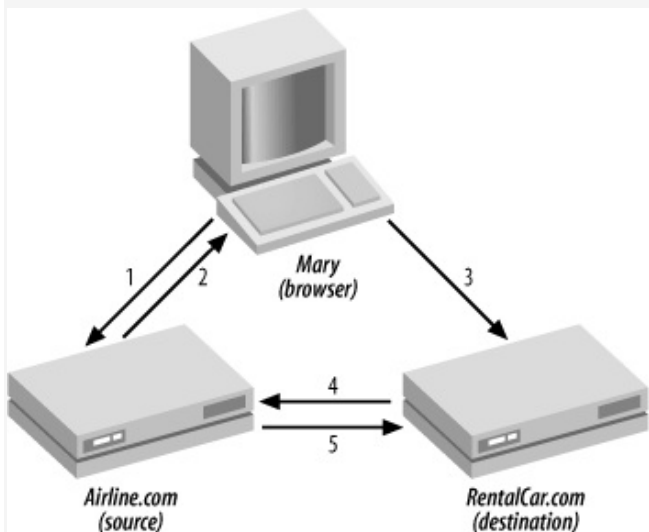


Figure 11-2. Browser pull profile

POST

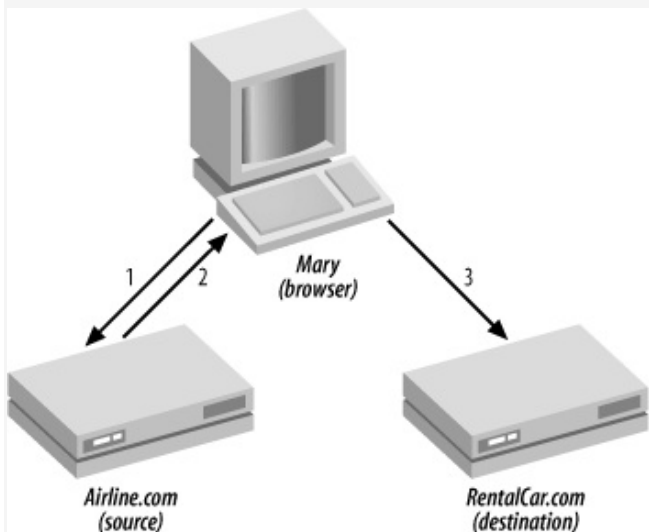


Figure 11-3. Browser push profile

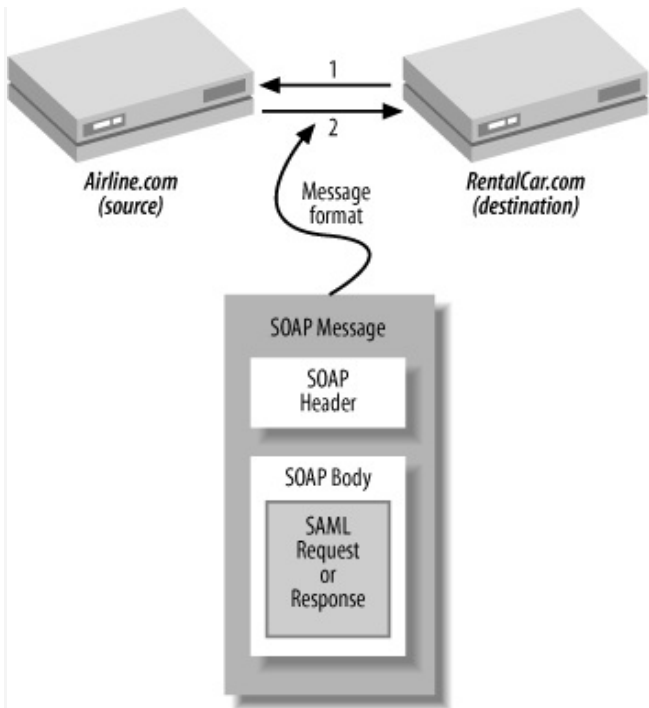


Figure 11-4. Direct request and response

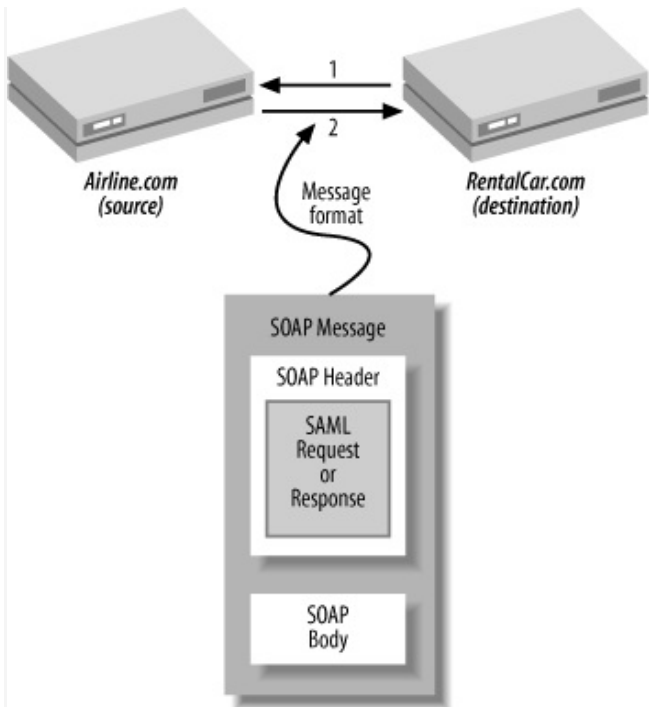


Figure 11-5. Web services profile

Identity Provisioning

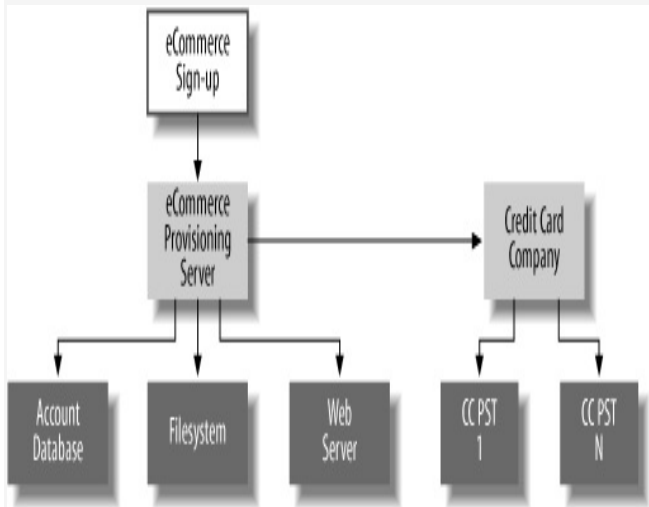


Figure 11-6. E-commerce provisioning

SPML Requests and Responses

```
<addRequest/>
```

<modifyRequest/>

<deleteRequest/>

<searchRequest/>

<schemaRequest/>

<addRequest/>

<addRequest>
 <attributes>
 <attr name="objectclass">

<value>urn:....:SPML:interop:intero

</attr>

```
<attr name="cn">
  <value>Jane Doe</value>
</attr>
<attr name="mail">
```

```
<value>Jane.Doe@windley.com</value>
```

```
  </attr>
  <attr name="description">
    <value>Jane Doe
interopUser Subscription</value>
  </attr>
  <attr name="memberLevel">
    <value>1</value>
  </attr>
  <attr name="company">
    <value>Jane Doe Supply
Co</value>
  </attr>
  <attr
name="registrationTime">
    <value>17-Nov-2004
12:00 </value>
```

```
        </attr>
    </attributes>
</addRequest>
```

```
    <addResponse result =
"urn:....:SPML:1:0#success">
        <identifier type=
"urn:....:SPML:1:0#EmailAddress">

<id>Jane.Doe@windley.com</id>
        </identifier>
        <attributes>
```



```
        <attr  
name="mailBoxLimit">  
        <value>50MB</value>  
    </attr>  
    </attributes>  
</addResponse>
```



Representing and Managing Authorization Policies

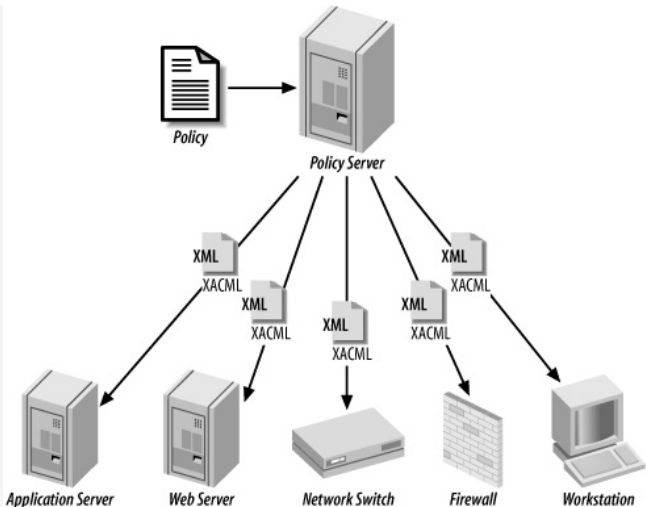


Figure 11-7. Using a policy server










```
<Request>
  <Subject/>
  <Resource>
    <Attribute
```

```
AttributeId="urn:oasis:...:resourc
id">
```

```
<AttributeValue>SomeServer</Attrib
```

```
    </Attribute>
```

```
</Resource>
```

```
<Action>
```

```
  <Attribute
```

```
AttributeId="ServerAction">
```

```
<AttributeValue>login</AttributeVa
```

```
    </Attribute>
```

</Action>
</Request>

<Policy/>

<Policy
 PolicyId="SamplePolicy"
 RuleCombiningAlgId=
 "urn:....:xacml:1.0:rule-

combining-algorithm:first-
applicable">

<Policy/>

<Target/>

<Rule/>

<Target/>

<Rule/>

permit

deny

<Target/>

<Target>

<Subjects>

<AnySubject/>

```
</Subjects>
<Resources>
  <Resource>
    <ResourceMatch
```

```
MatchId="urn:...:xacml:1.0:func
equal">
```

```
  <AttributeValue>
    SampleServer
  </AttributeValue>
```

```
<ResourceAttributeDesignator
```

```
AttributeId="urn:...:xacml:1.0:res
id"/>
```

```
  </ResourceMatch>
</Resource>
</Resources>
<Actions>
  <AnyAction/>
</Actions>
</Target>
```

<Target/>

<Policy/>

<Rules/>

<Rule/>

Permit

<Rule/>

```
<Rule RuleId="LoginRule"  
Effect="Permit">
```

```
<Target/>
```

```
<Target>
  <Subjects>
    <AnySubject/>
  </Subjects>
  <Resources>
    <AnyResource/>
  </Resources>
  <Actions>
    <Action>
      <ActionMatch
```

```
MatchId="urn:...:xacml:1.0:function
equal">
```

```
  <AttributeValue
    Login
  </AttributeValue>
```

```
<ActionAttributeDesignator
AttributeId="ServerAction"/>
```



```
    </ActionMatch>  
    </Action>  
  </Actions>  
</Target>
```

```
<Rule/>
```

```
        <Condition
FunctionId="urn:...:xacml:1.0:func

        <Apply
        FunctionId=

"urn:...:xacml:1.0:function:time-
greater-than-or-equal">
        <Apply

FunctionId="urn:...:xacml:1.0:func
one-and-only">

<EnvironmentAttributeDesignator
        AttributeId=

"urn:...:xacml:1.0:environment:cur
time"/>
        </Apply>
        <AttributeValue>
        09:00:00
        </AttributeValue>
</Apply>
```

```
<Apply  
  FunctionId=
```

```
"urn:...:xacml:1.0:function:time-  
less-than-or-equal">
```

```
<Apply
```

```
FunctionId="urn:...:xacml:1.0:func  
one-and-only">
```

```
<EnvironmentAttributeDesignator
```

```
DataType="http://www.w3.org/2001/X
```

```
  AttributeId=
```

```
"urn:...:xacml:1.0:environment:cur  
time"/>
```

```
</Apply>
```

```
<AttributeValue>
```

```
  17:00:00
```

```
</AttributeValue>
```

```
</Apply>
```

</Condition>

```
($ENV(current_time) >=
"09:00:00")
    AND ($ENV(current_time) <=
"17:00:00")
```

Deny

```
<!-- A final, "fall-through"  
Rule that always Denies -->  
  <Rule RuleId="FinalRule"  
Effect="Deny"/>
```

Permit

<Response>

<Result>

<Decision>Permit</Decision>

<Status>

<StatusCode

Value="urn:....:xacml:1.0:status:ok

</Status>

</Result>

</Response>



Conclusion

Chapter 12. Fed Identity

Centralized Versus Federated Identity

The Mirage of Centralized Efficiency

Network Effects and Digital Identity Management







Federation in the Credit Card Industry

Benefits of Federated Identity

Digital Identity Standards

Microsoft, IBM, and the WS-* Roadmap

Microsoft/IBM

Oasis

Liberty Alliance

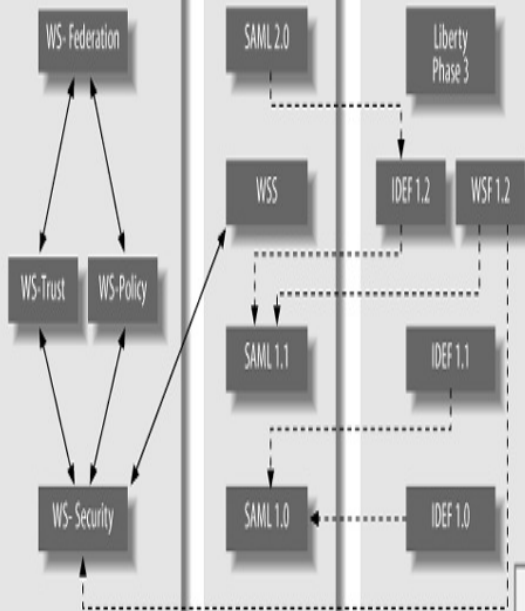


Figure 12-1. The interrelationship among federation standards

OASIS

Liberty Alliance

Internet2 and Shibboleth

The Future of Federation Standards

Three Federation Patterns

Pattern 1: Ad Hoc Federation

Pattern 2: Hub-and-Spoke Federation

Bank of America: a cautionary tale

Scenario 3: Identity Network





Addressing the Problem of Trust

**A Secure,
Protected
Environment**

The Future of Federated Identity Networks

Conclusion

Chapter 13. An Architecture for Digital Identity

Identity

Management Architecture





The Benefits of an Identity Management Architecture

Success Factors











Roadblocks

Identity Management Architecture Components

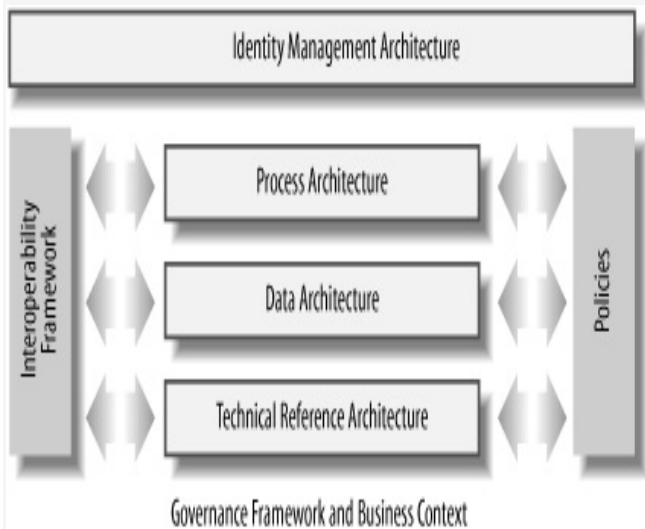


Figure 13-1. Components in an identity management architecture

Conclusion

Chapter 14. Gov and Business Modeling







IMA Lifecycle

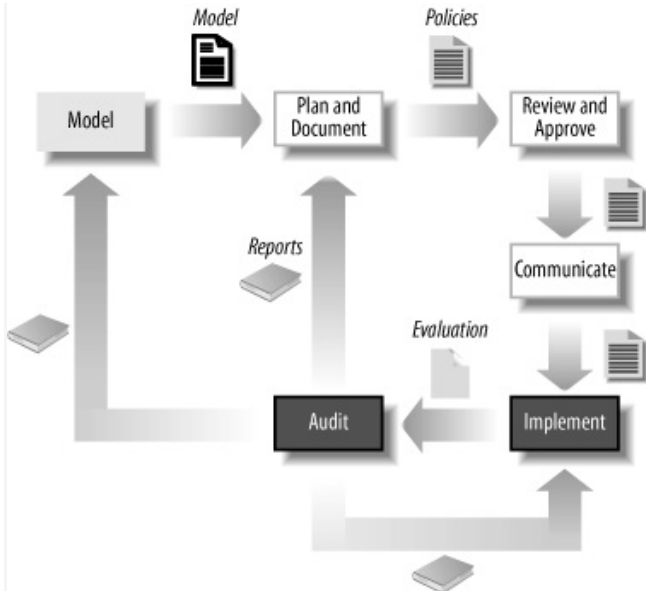


Figure 14-1. Identity management architecture lifecycle

IMA

Governance Model

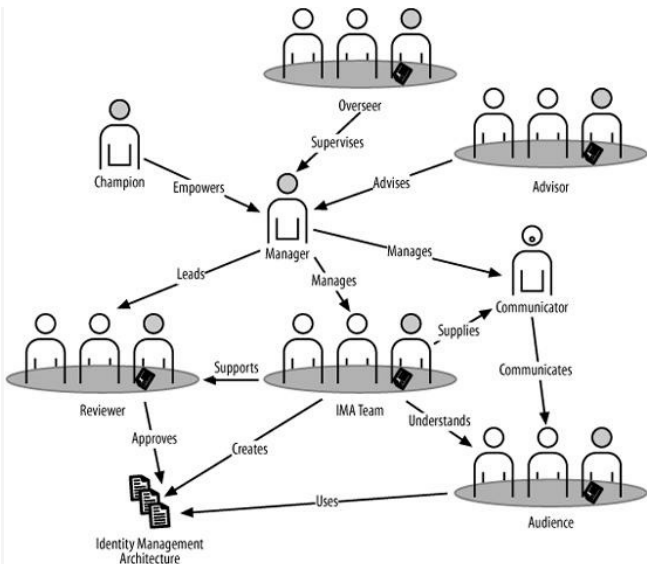


Figure 14-2. Illustrative IMA governance model





Initial Steps

Creating a Vision











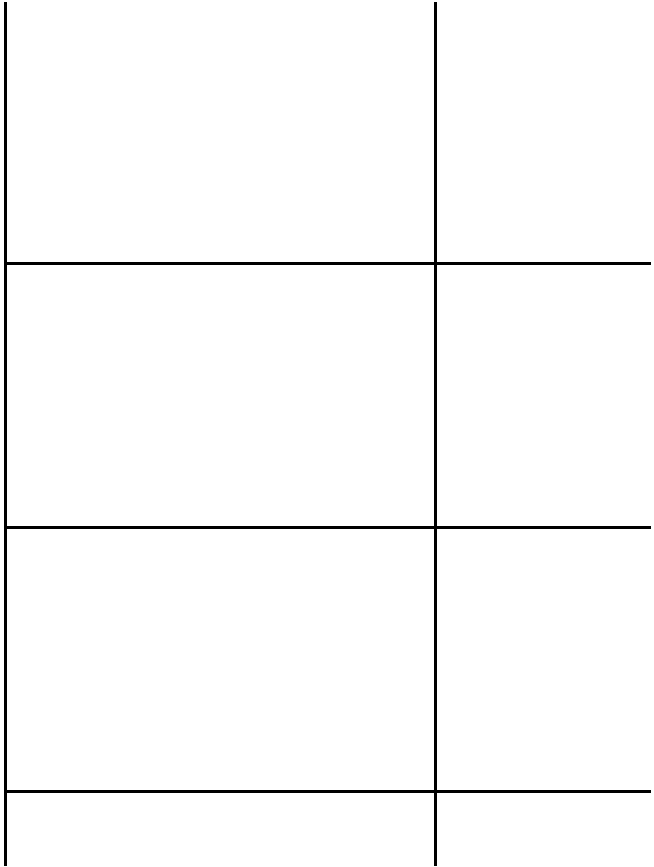




IMA Governing Roles

Table 14-1. IMA governance roles

--	--



Primary Roles















Supporting Roles



Resources

What to Outsource

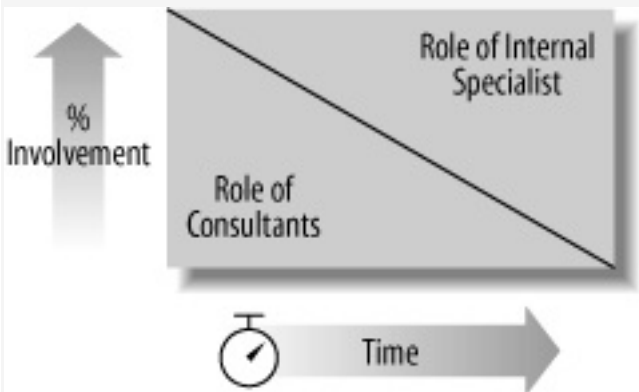


Figure 14-3. Moving from external consultants to internal specialists over time

Understanding the Business Context

Business Function Matrix











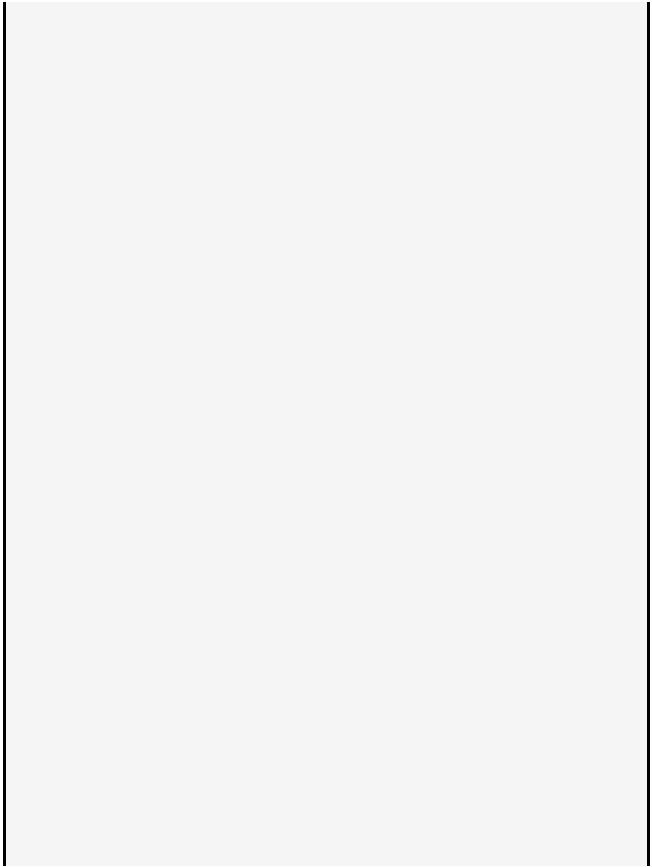


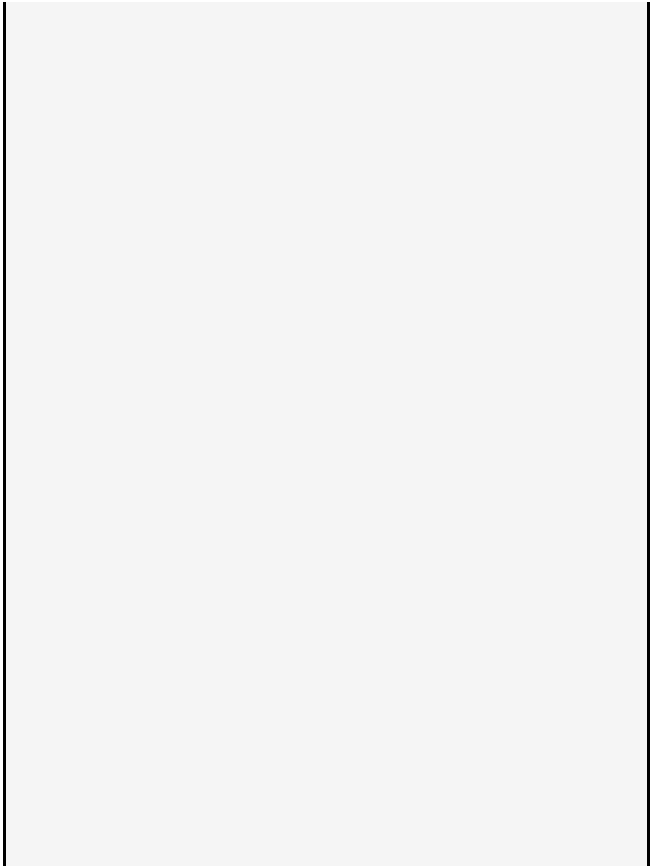
Creating the Business Function Matrix

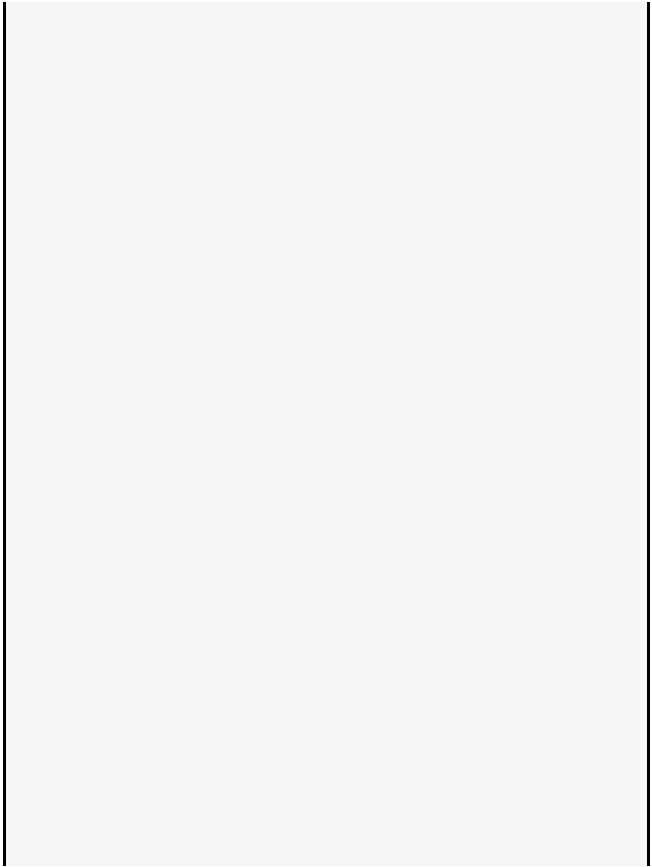


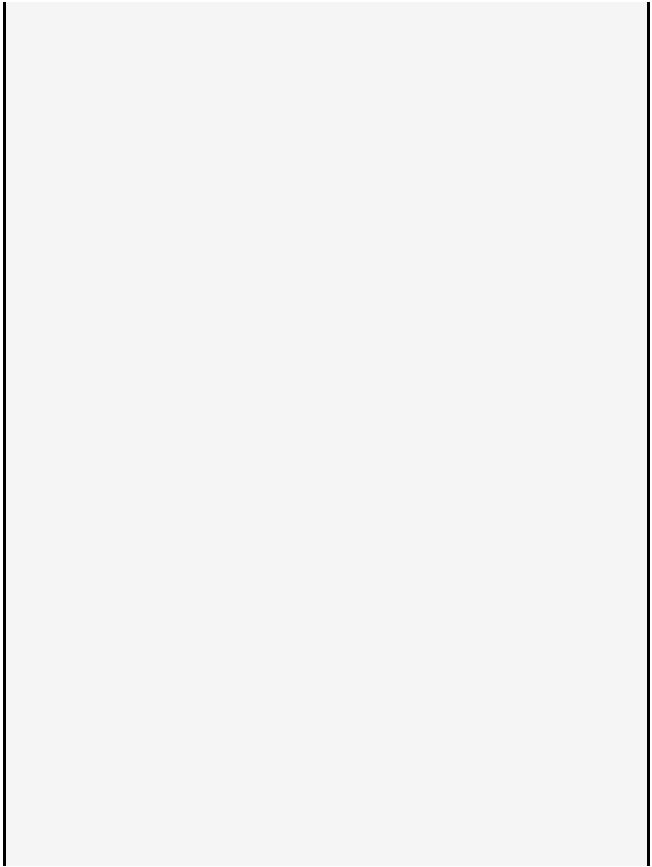
IMA Principles

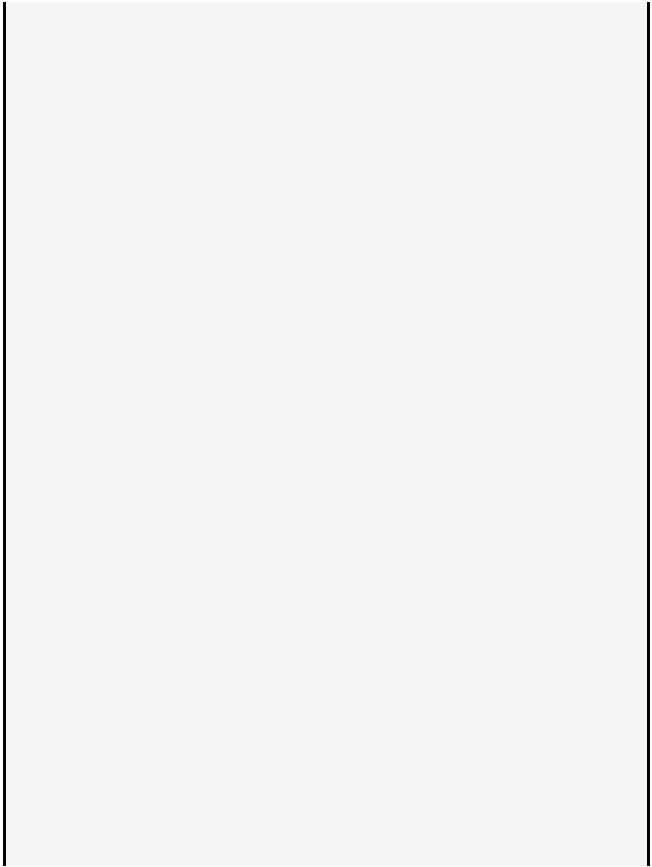
Principles for E-government in Utah

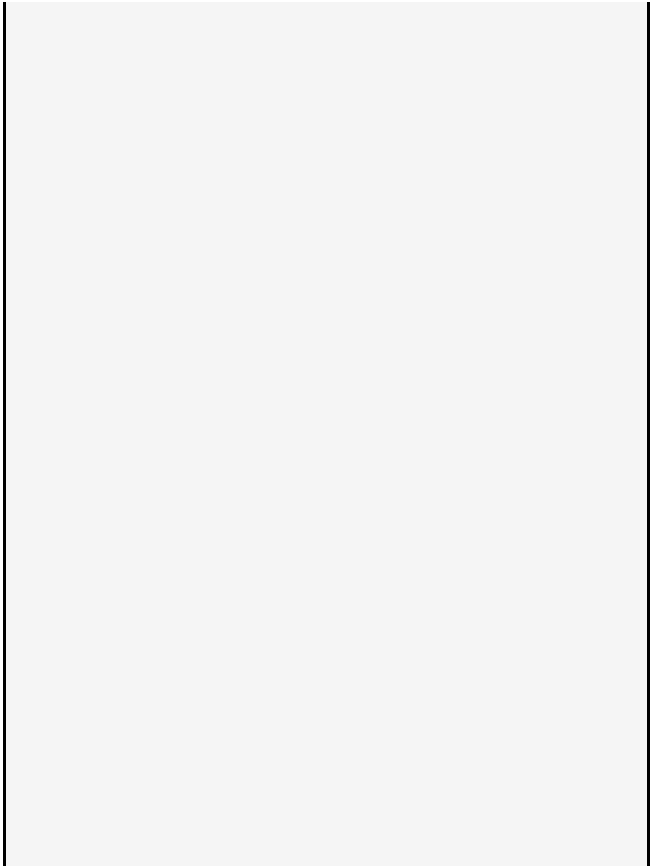


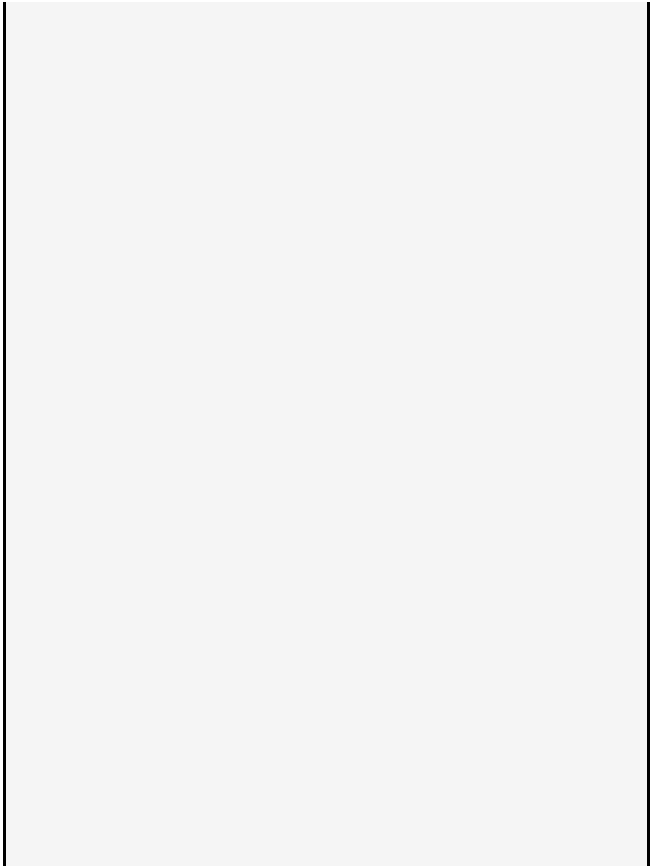


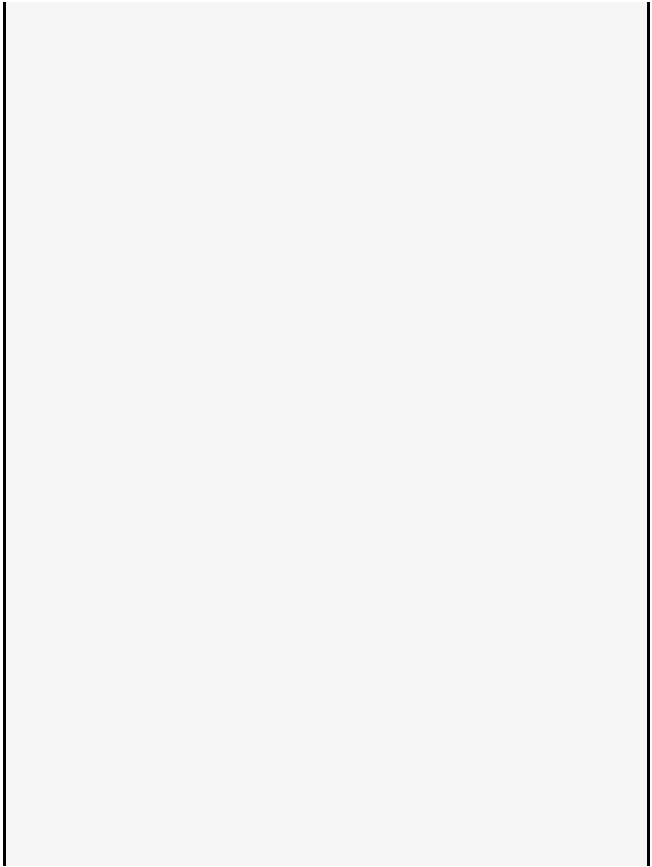


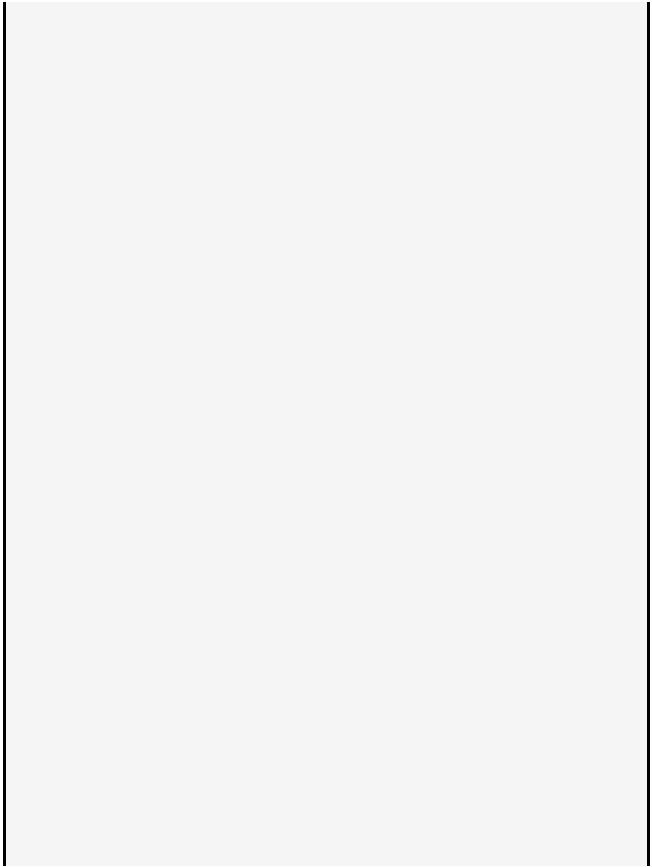












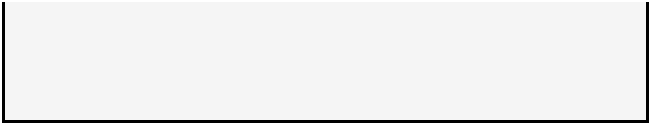
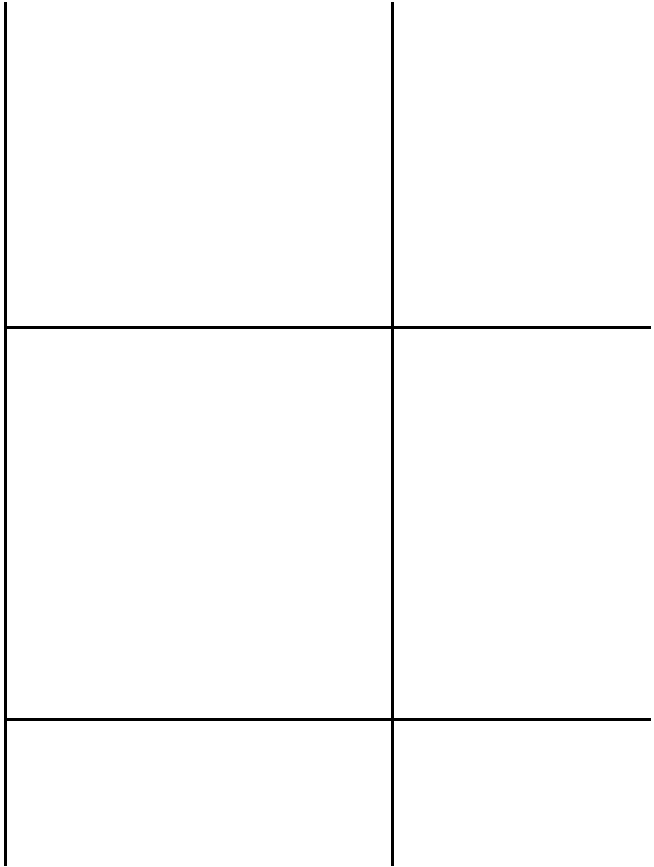
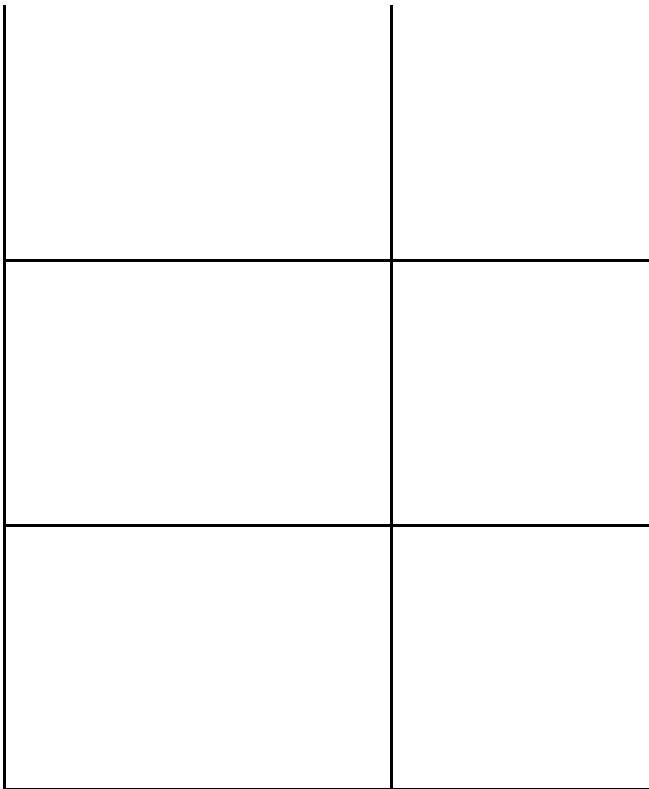


Table 14-2. IMA principles





Conclusion

Chapter 15. Identifier Maturity Models and Process Architectures

Maturity Levels

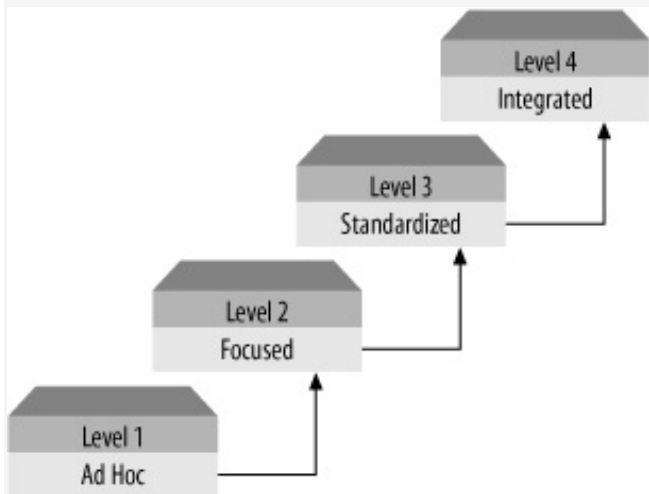
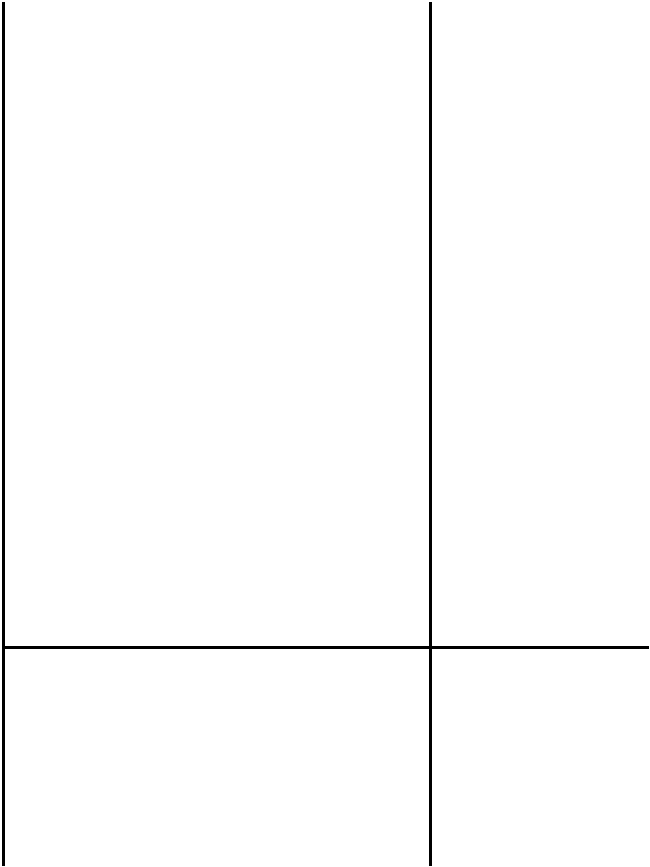


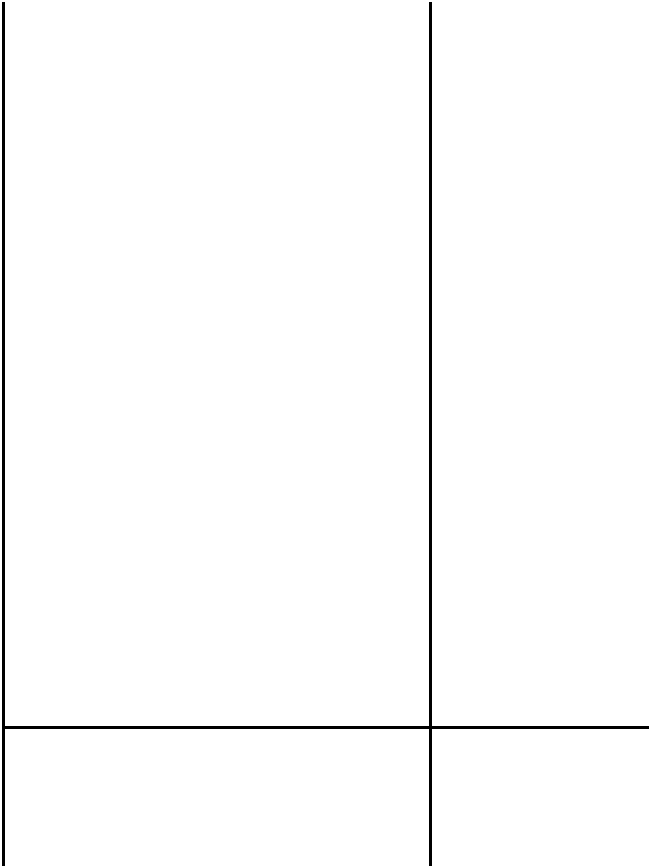
Figure 15-1. Levels of digital identity maturity

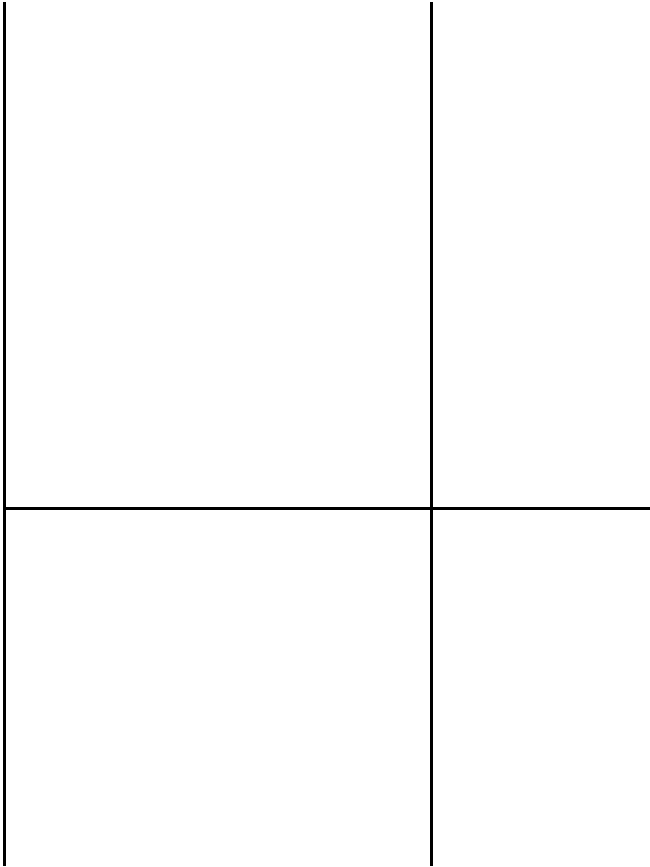
The Maturity Model

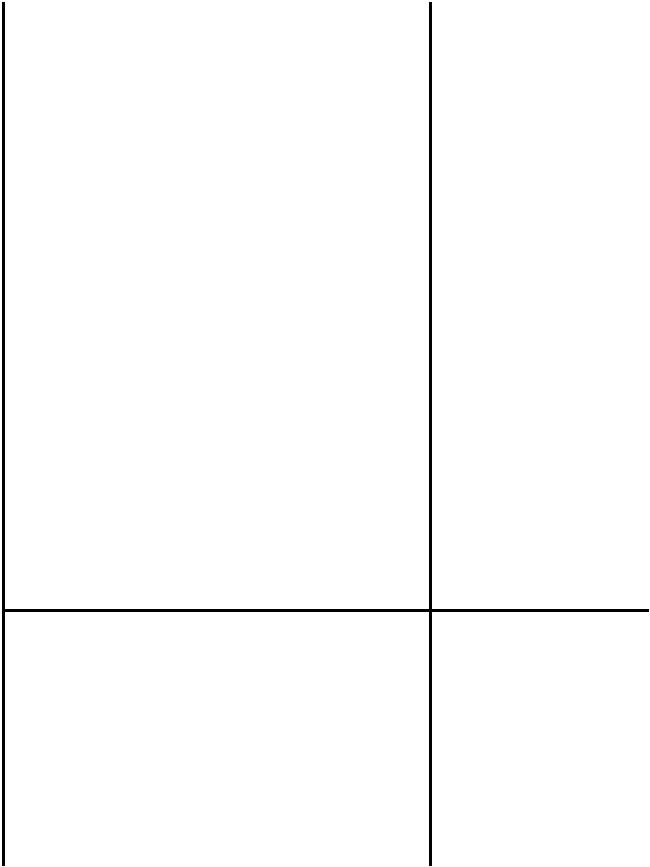
Level 1: Ad Hoc

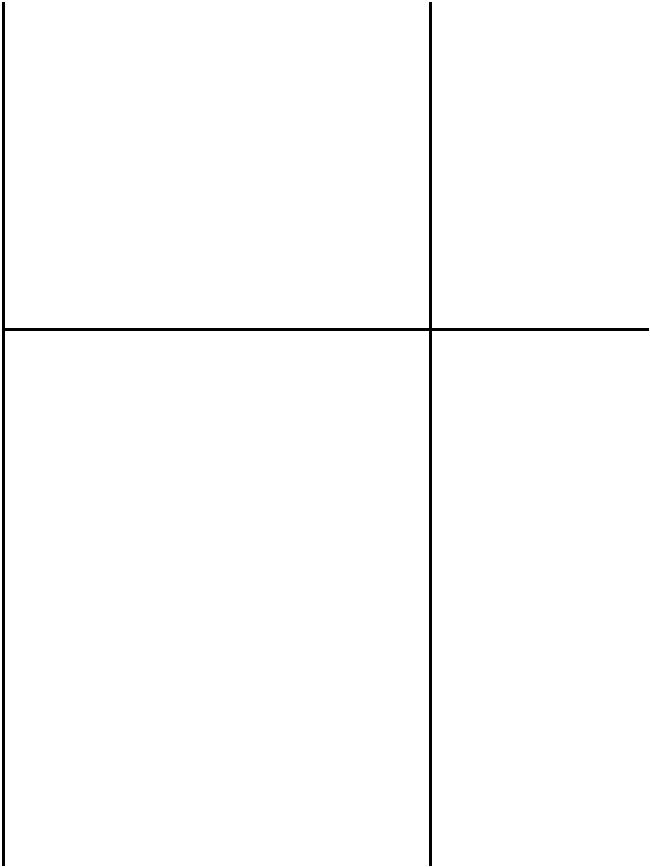
Table 15-
1. Characteristics of
level 1 identity
management practices

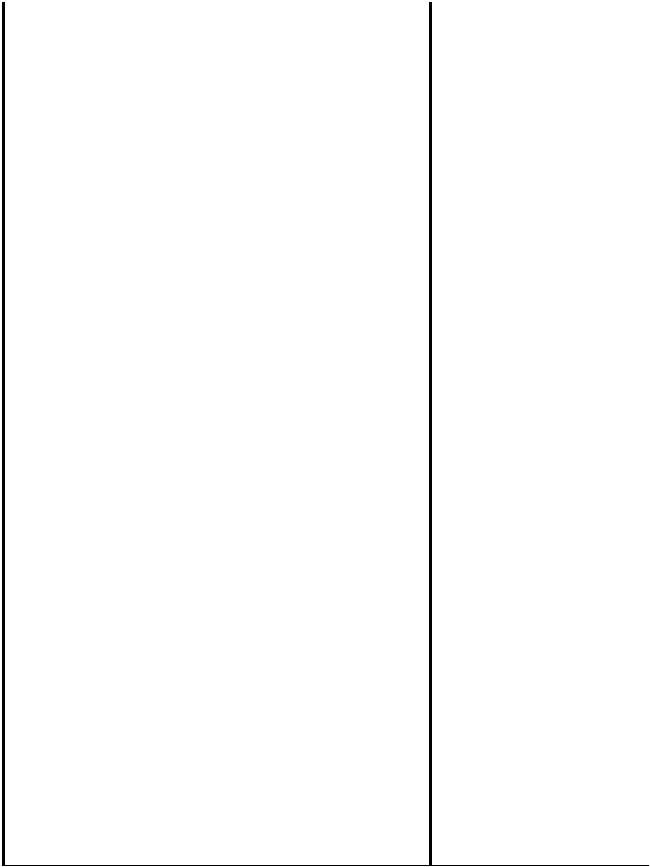


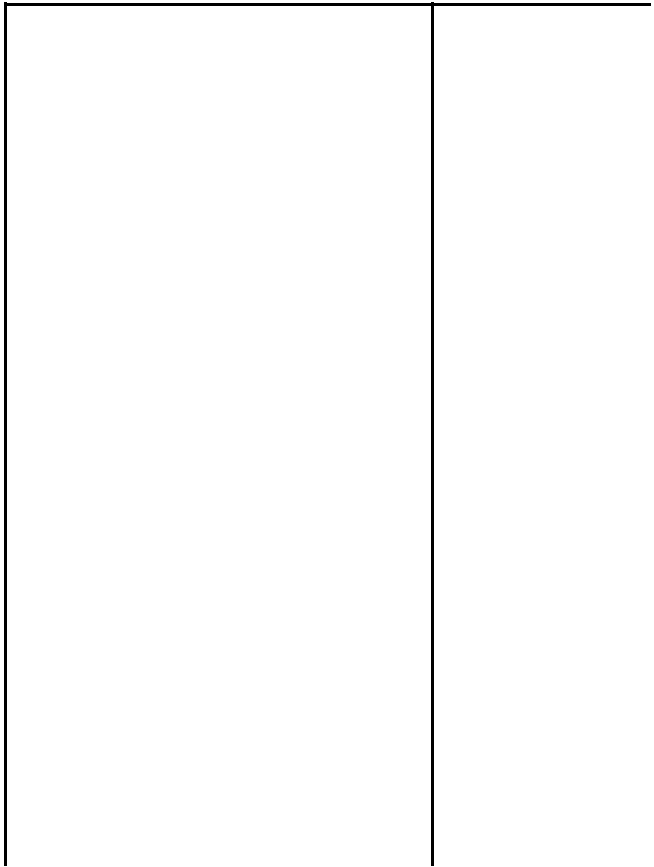


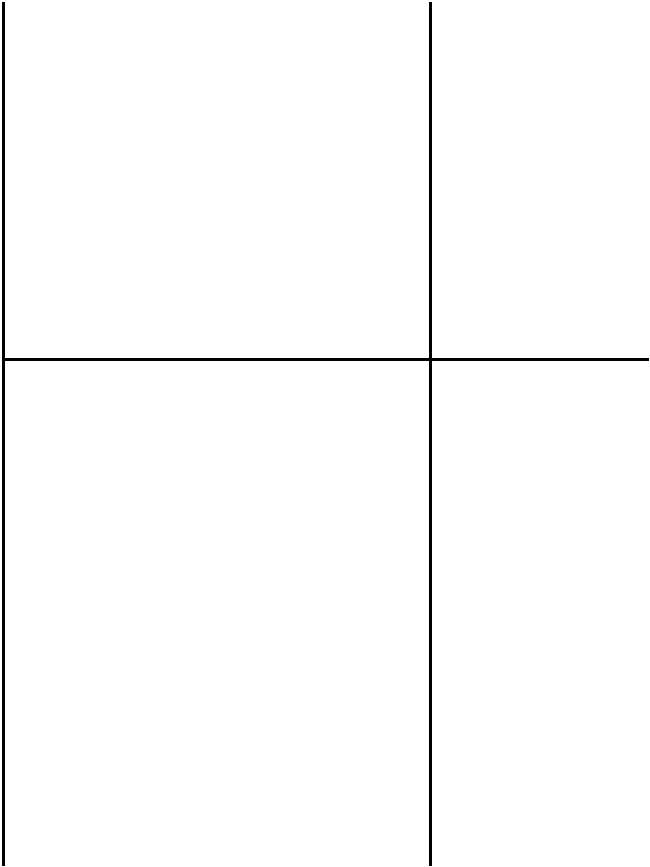


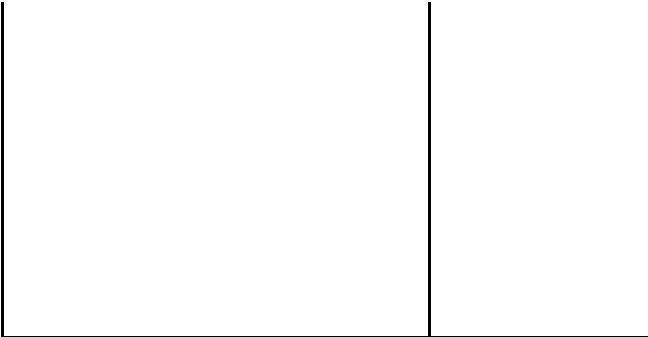






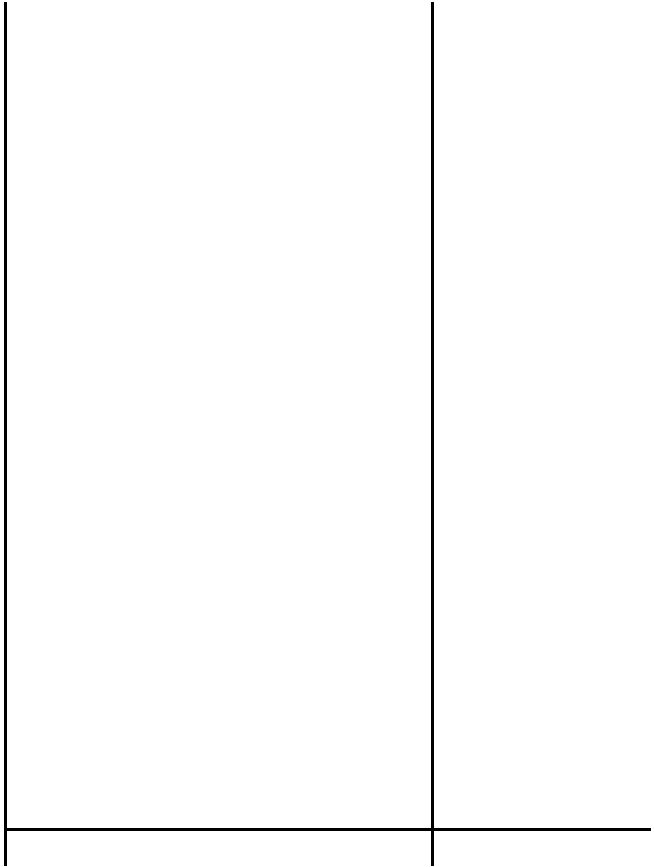


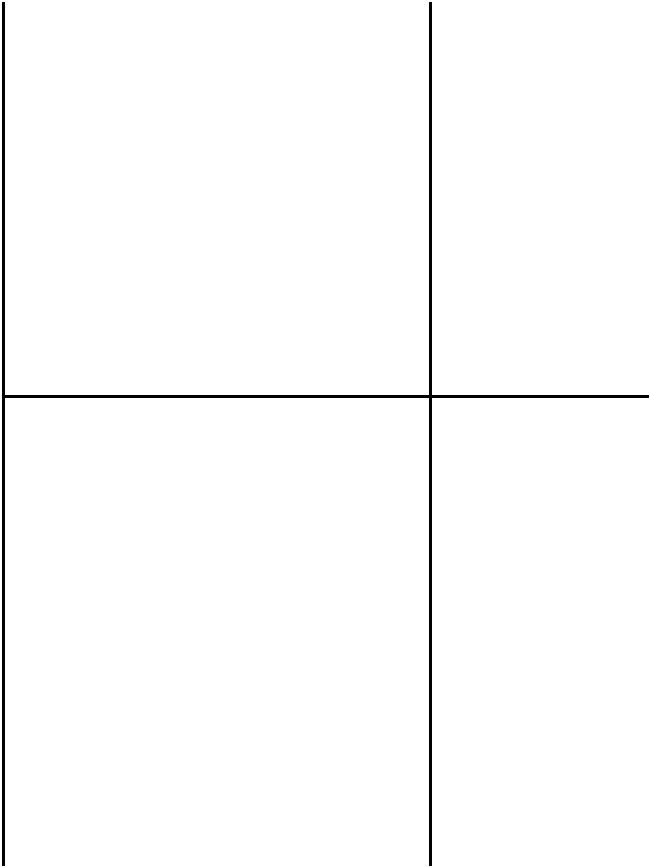


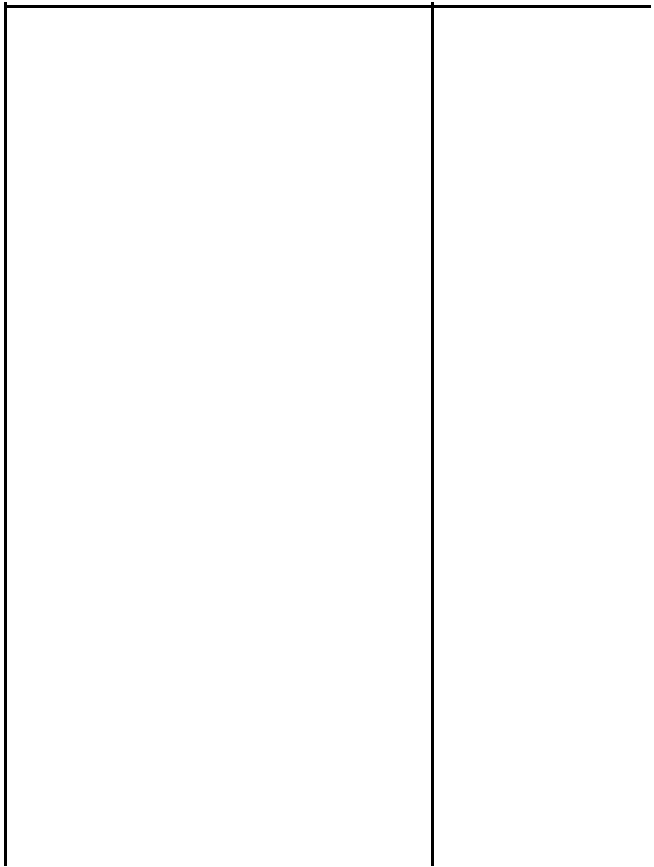


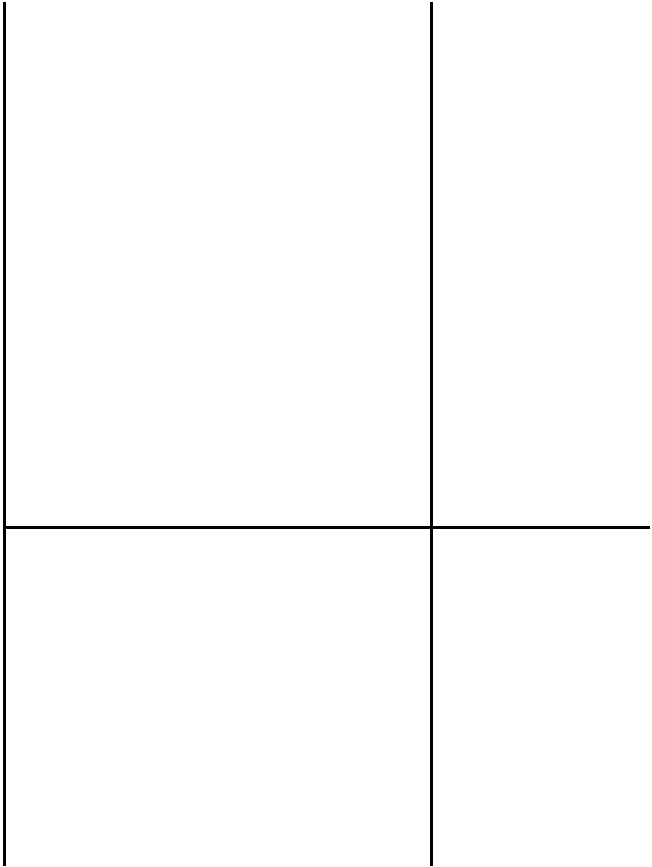
Level 2: Focused

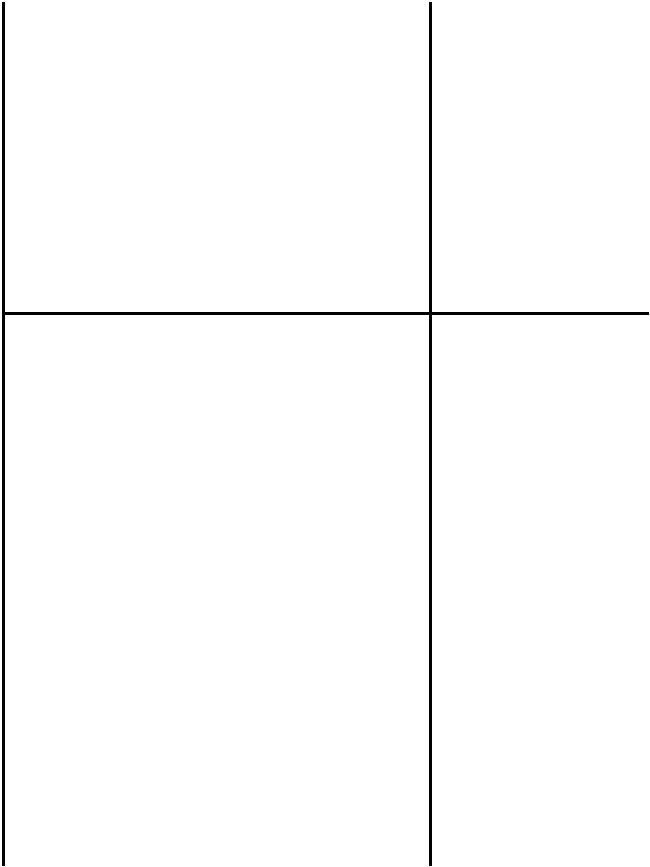
Table 15-
2. Characteristics of
level 2 identity
management practices

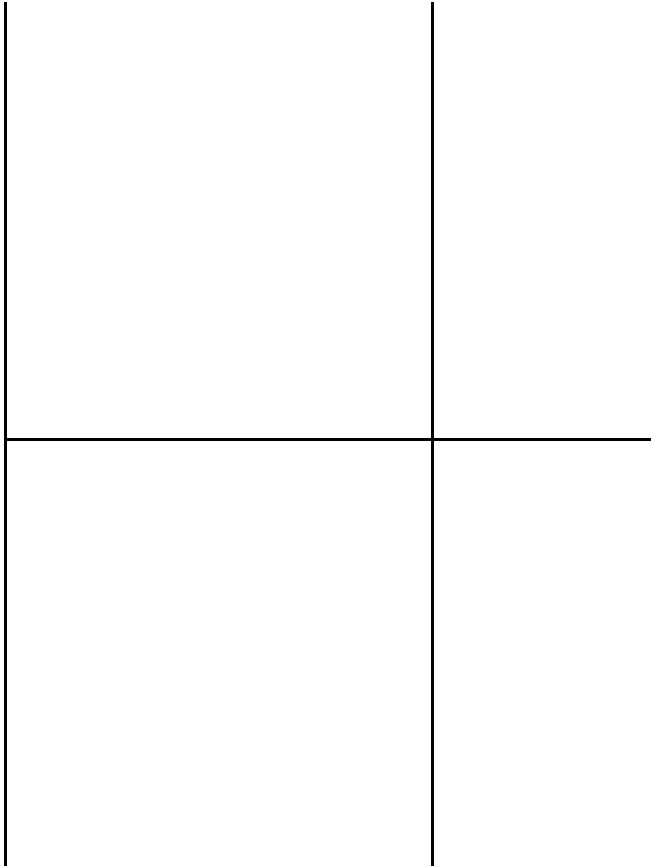


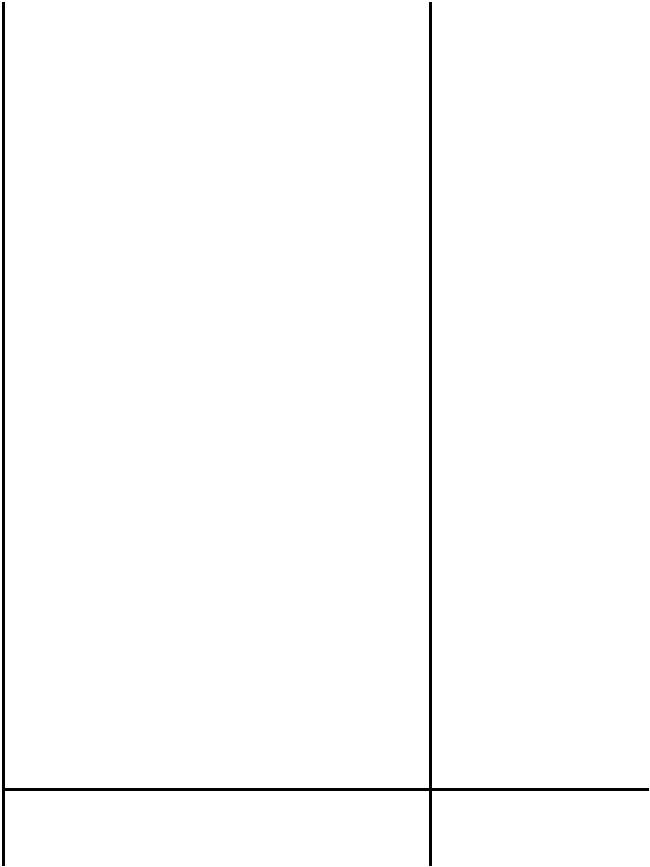


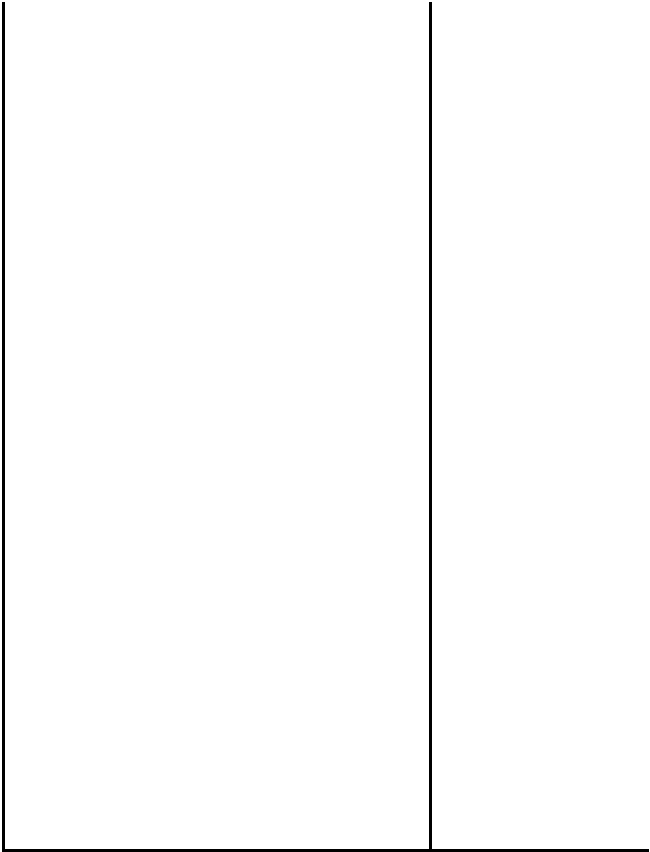








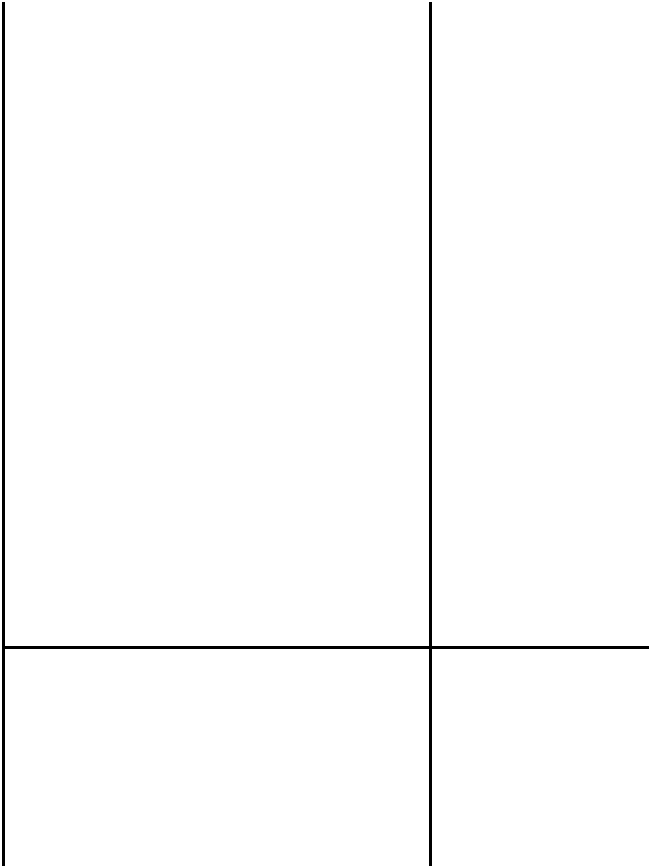


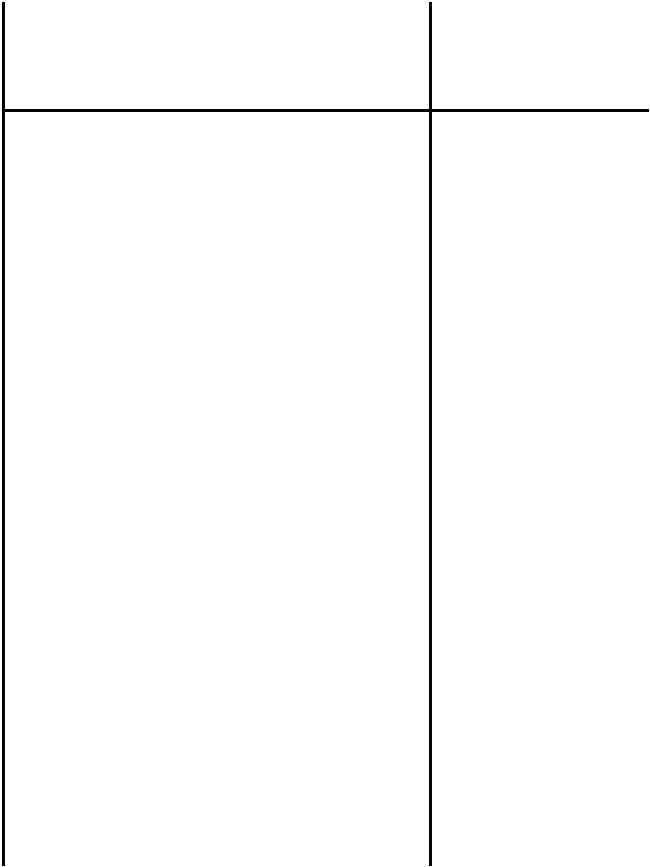


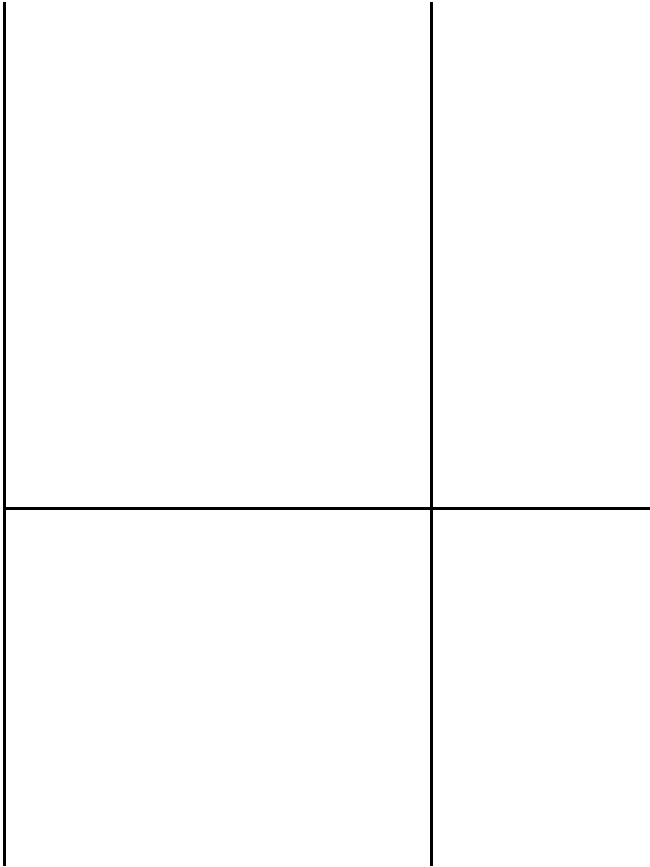
Level 3:

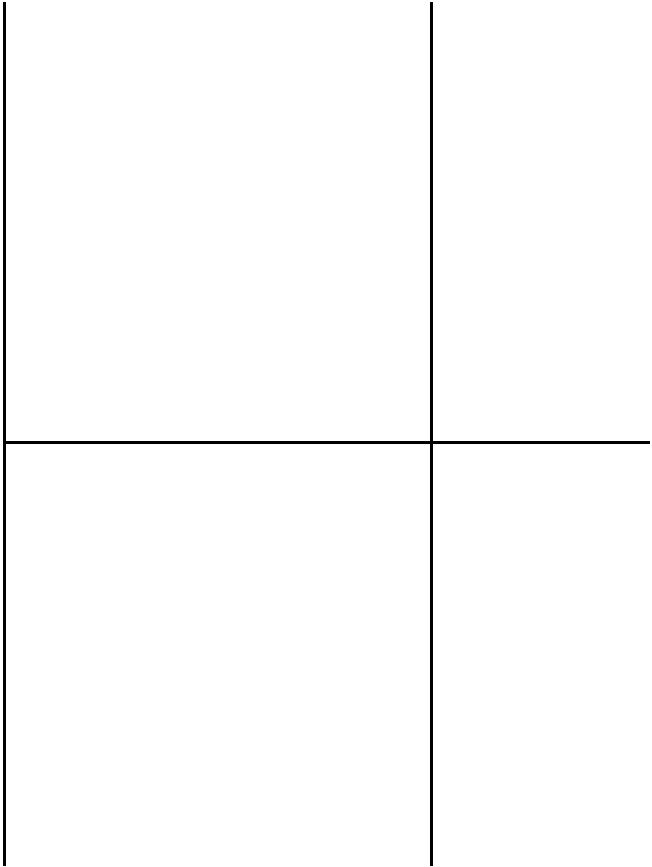
Standardized

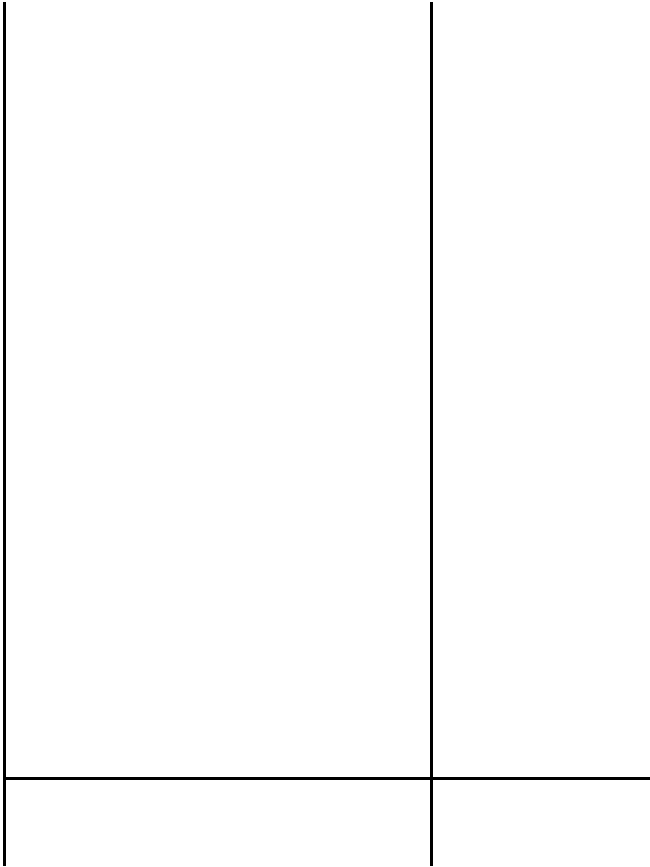
Table 15-
3. Characteristics of
level 3 identity
management practices

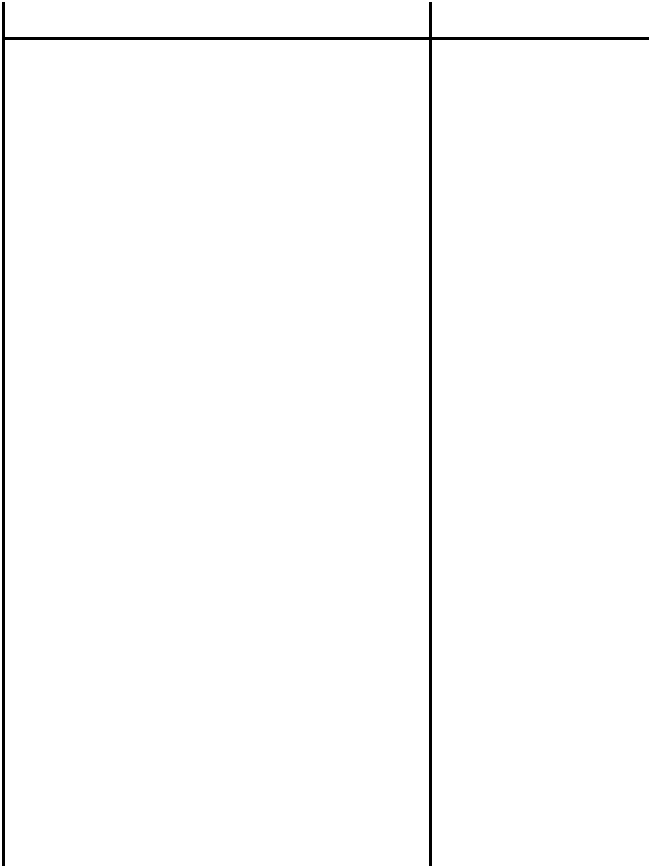


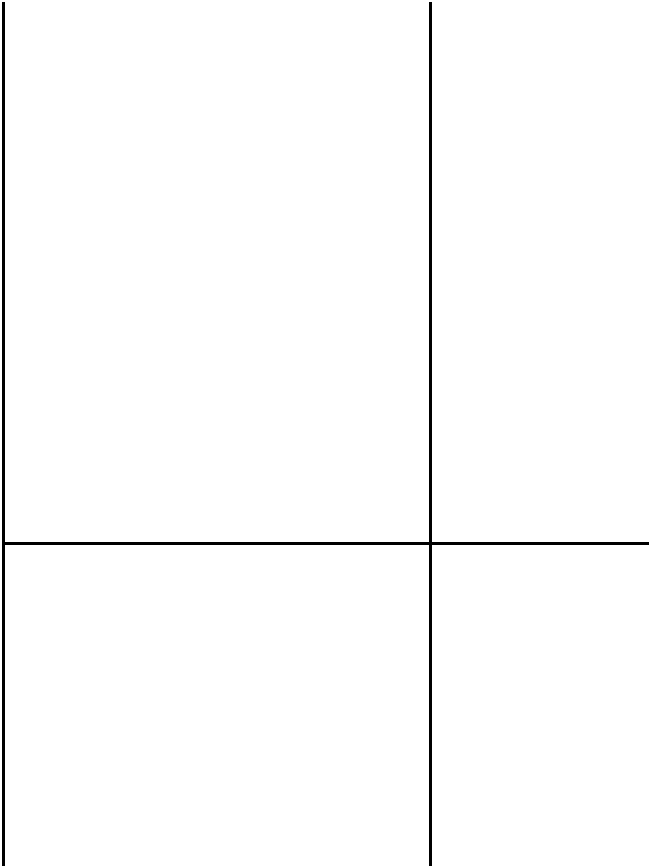


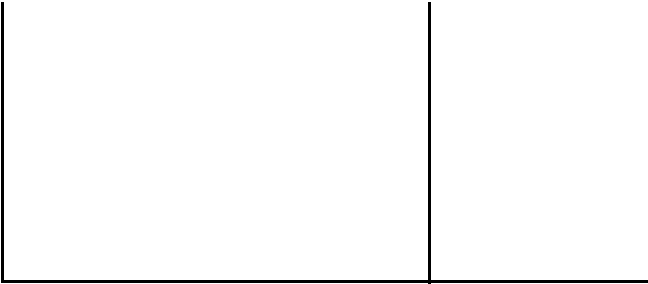








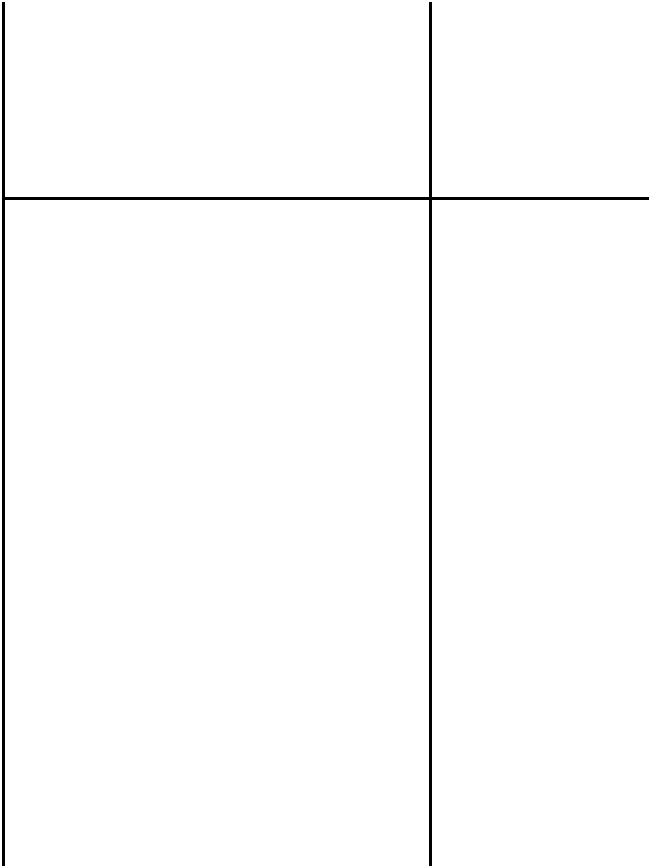


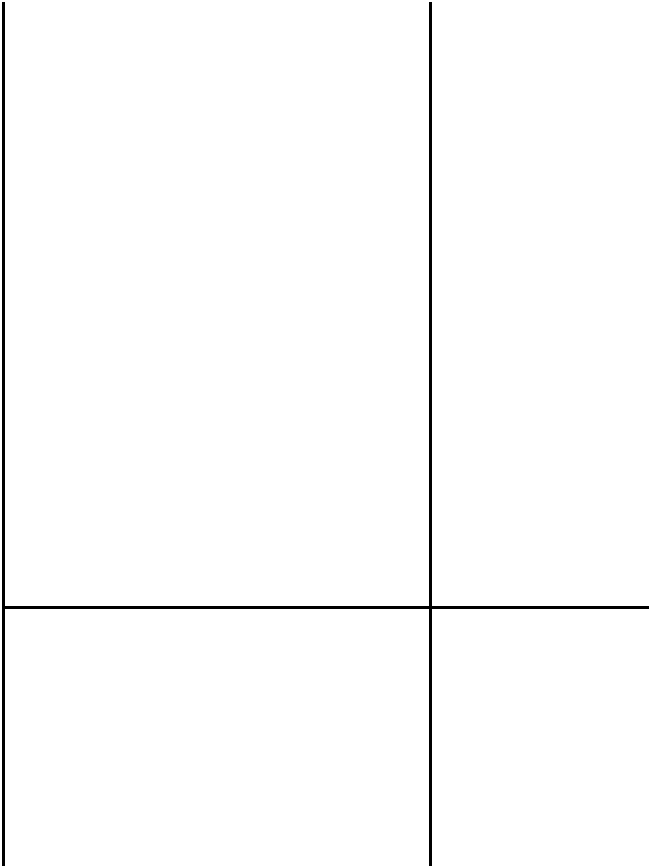


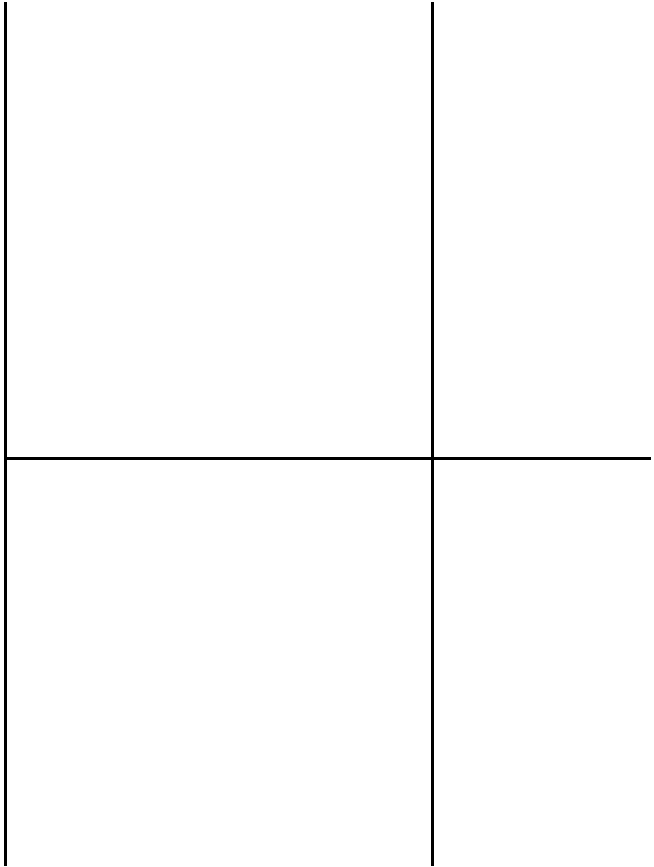
Level 4: Integrated

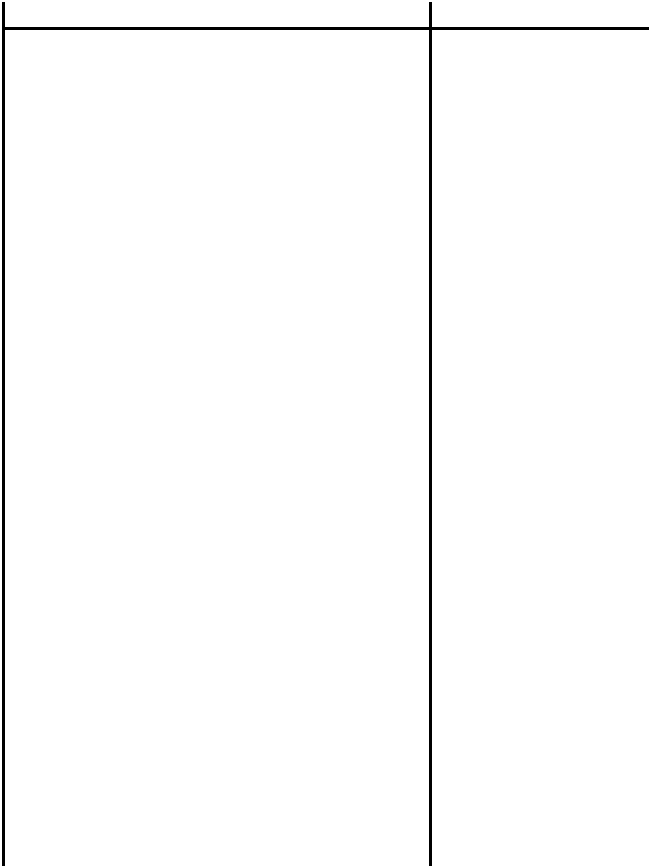
Table 15-
4. Characteristics of
level 4 identity

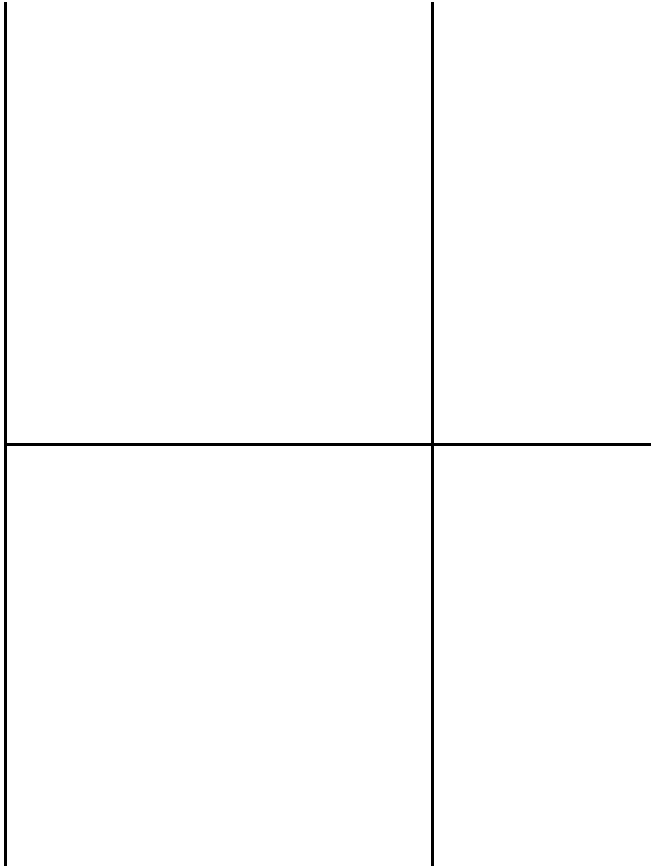
management practices

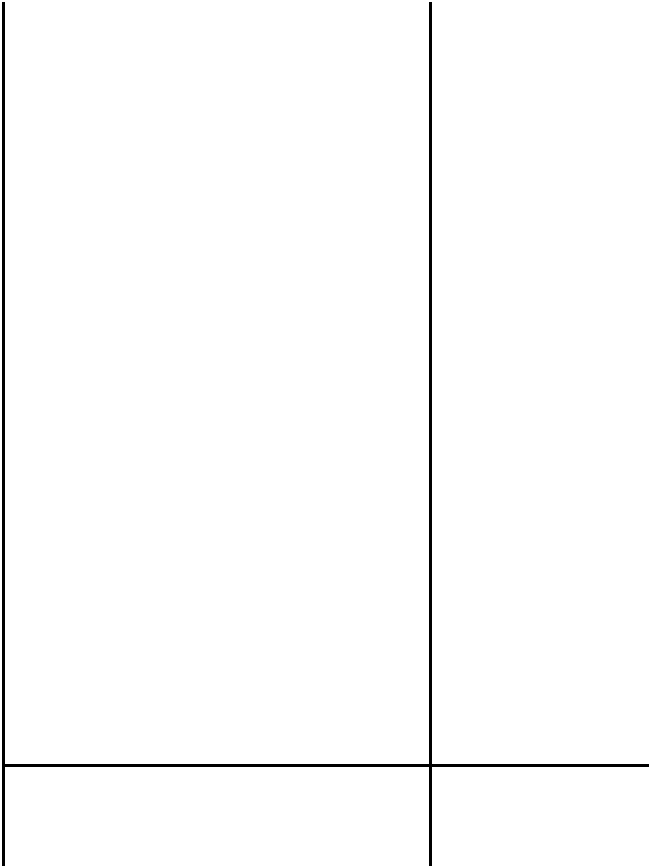


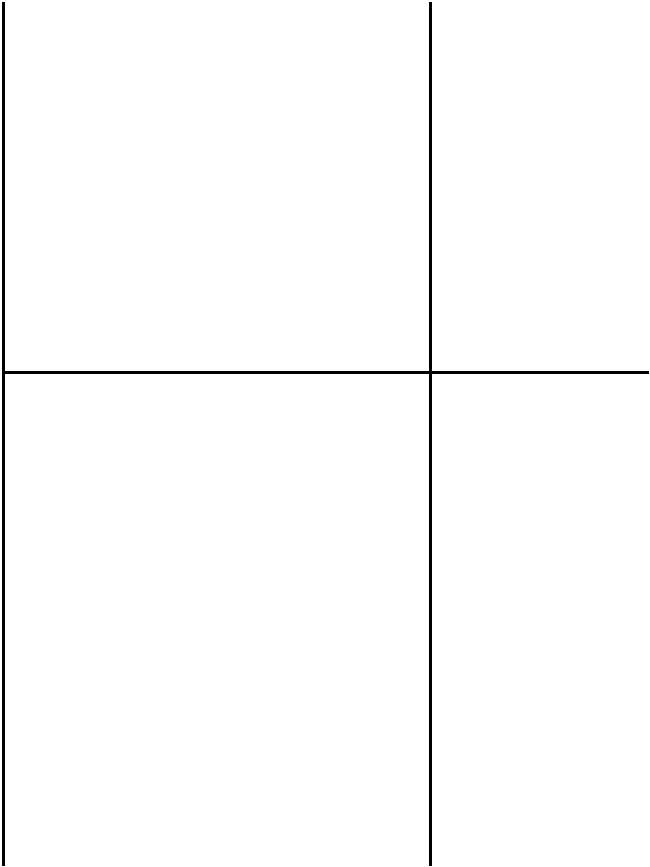


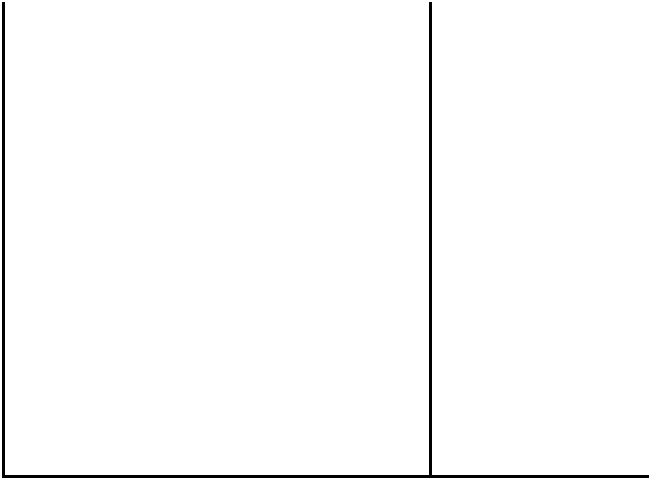












The Rights Steps at the Right Time

Finding Identity Processes

Evaluating Processes

A Practical Action Plan

Filling the Gaps with Best Practices

Conclusion

Chapter 16. Identifier Data Architectures

Build a Data

Architecture

Categorizing Identity Data

Baseline Identity Data Inventory
Process Matrix
Resource Mapping

Structure Identity Data

Data Element
Meta Data
Type and Format Standards

Exchanging Identity Data

Data Packages
Schema

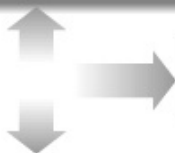


Figure 16-1. Components of a data architecture for identity

Processes Trump Data

Processes Link Identities

Employee Provisioning

















The Identity Data Inventory

Data Categorization

Identity Data Audit



























Identity Mapping





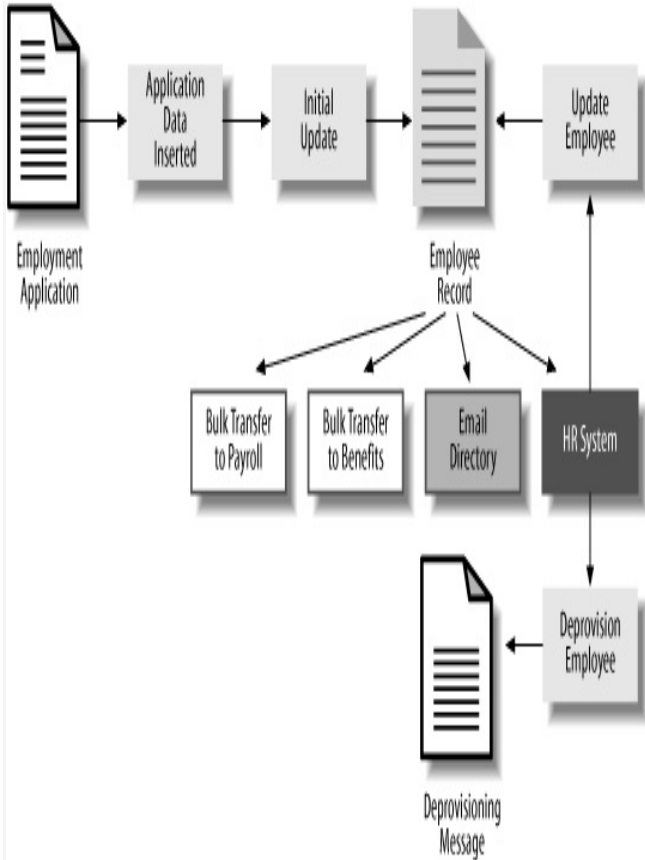


Figure 16-2. Example identity map for an employee record

Process-to-Identity Matrix

Table 16-1. Process-to-identity matrix for employee records





Identity Data Structure and Metadata







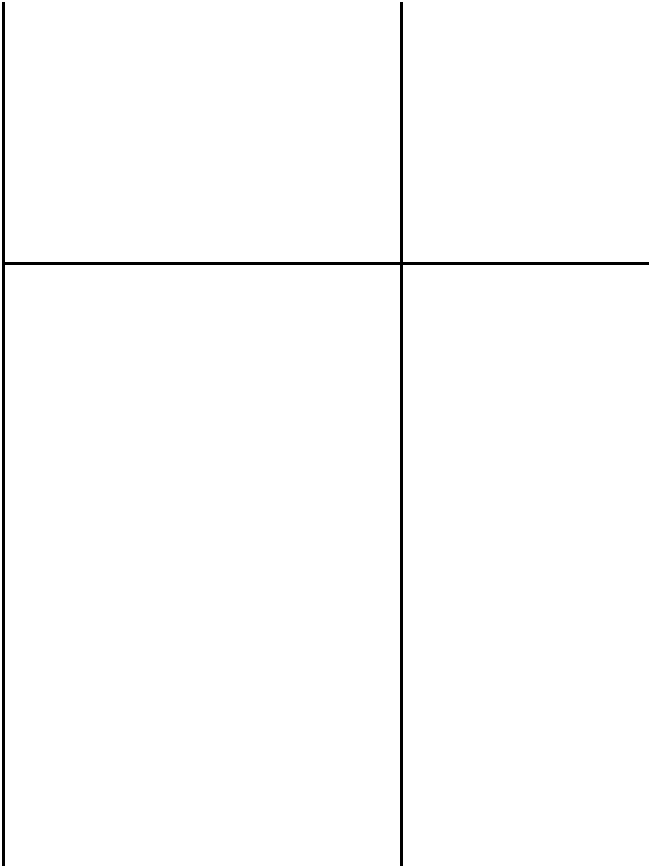
lname

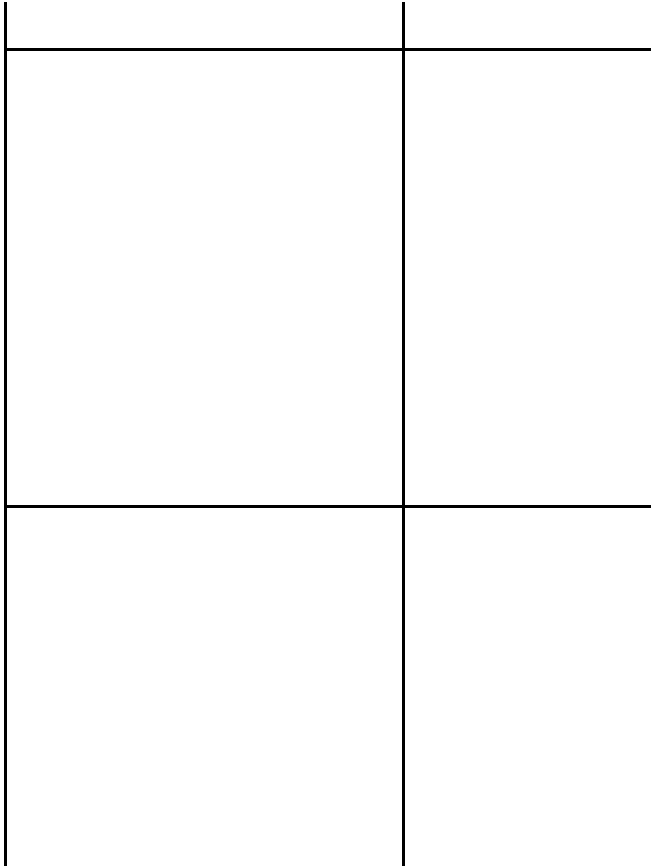
lastname

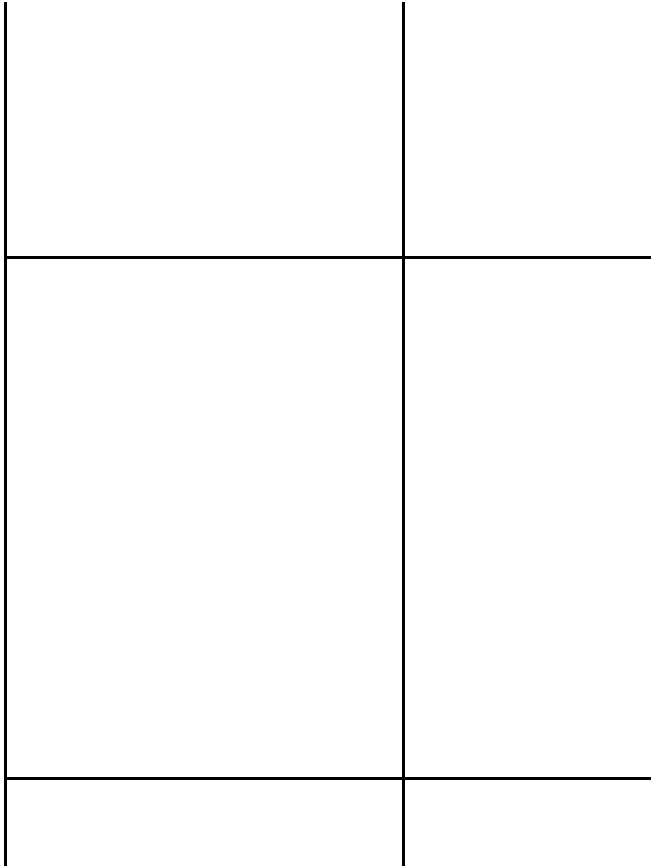


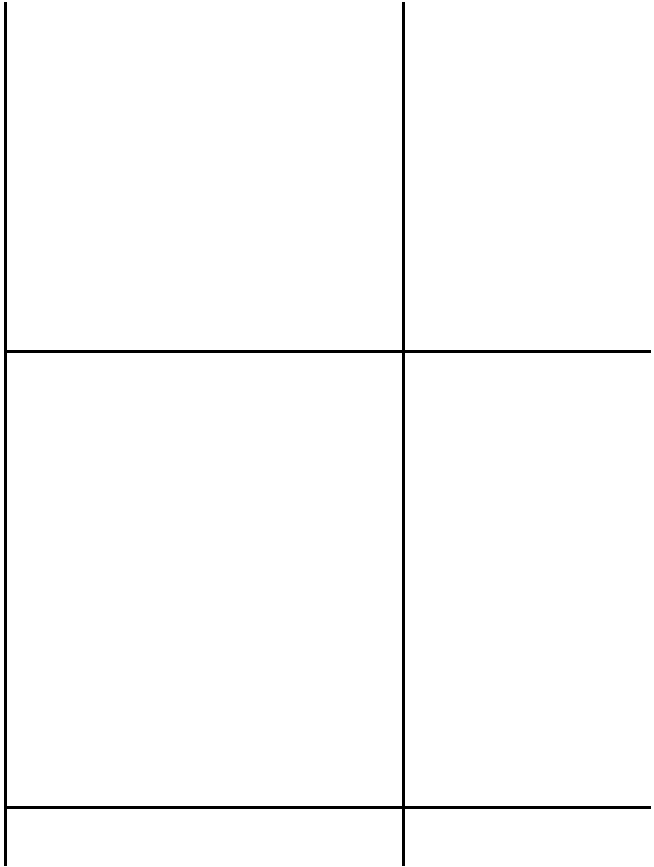
phone

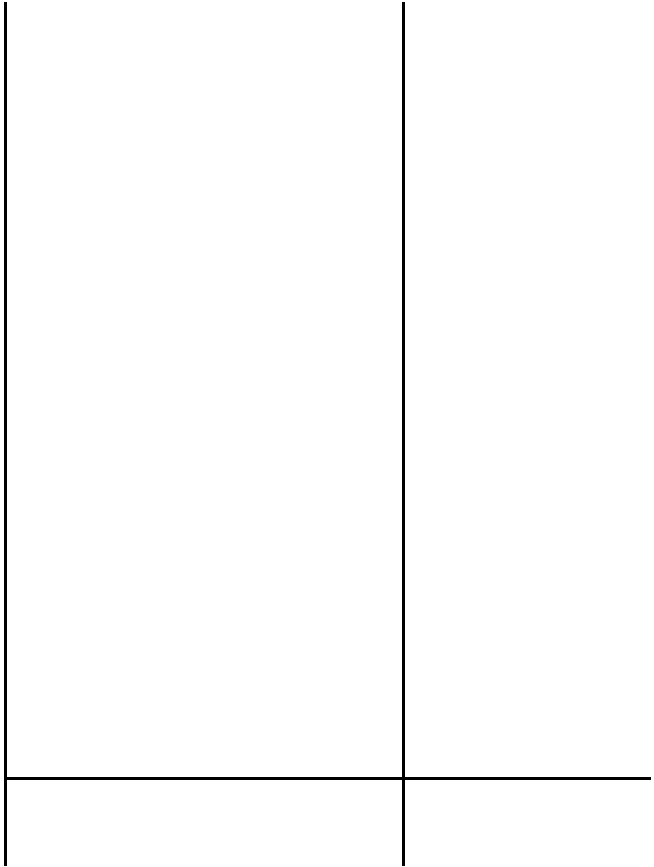
Table 16-2. Metadata for identity data elements

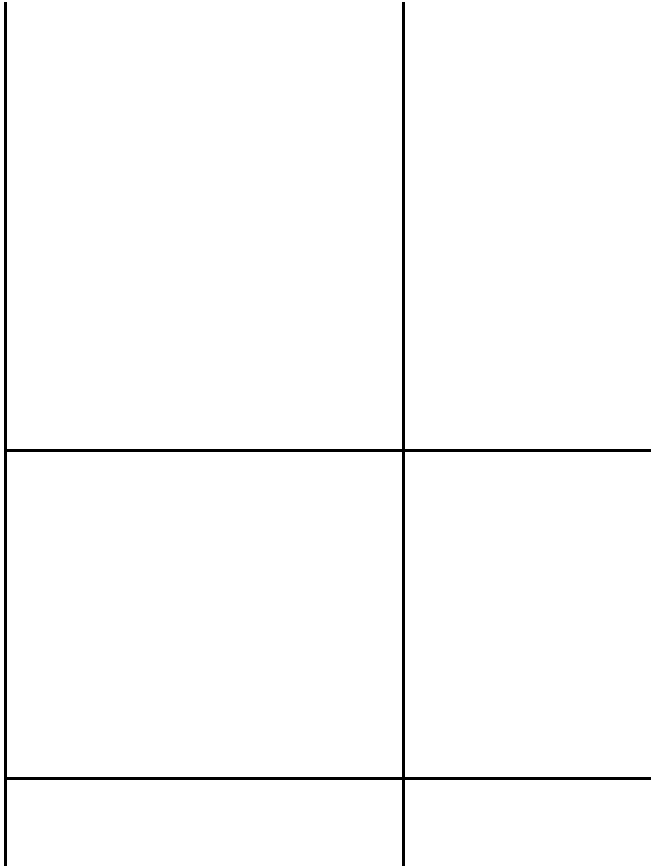


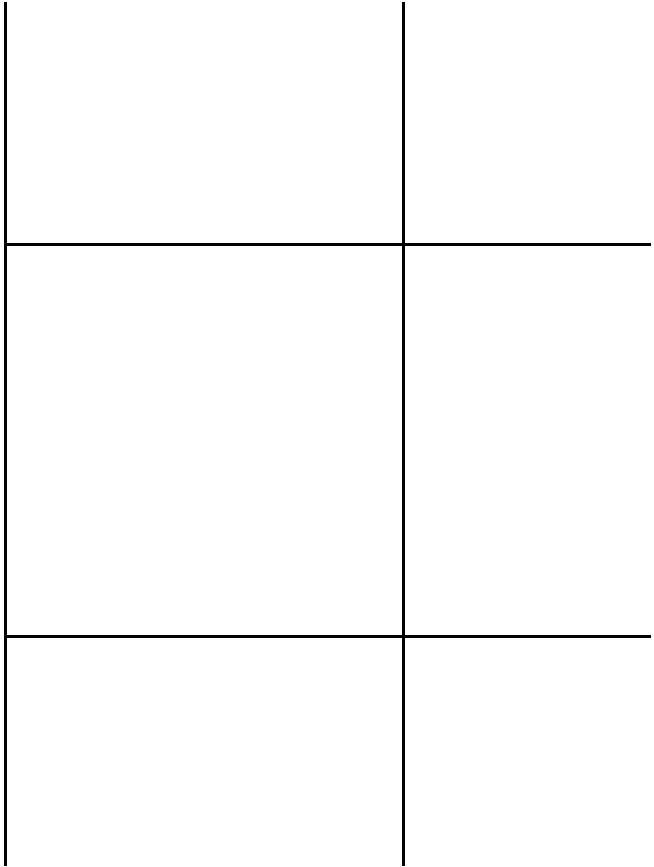


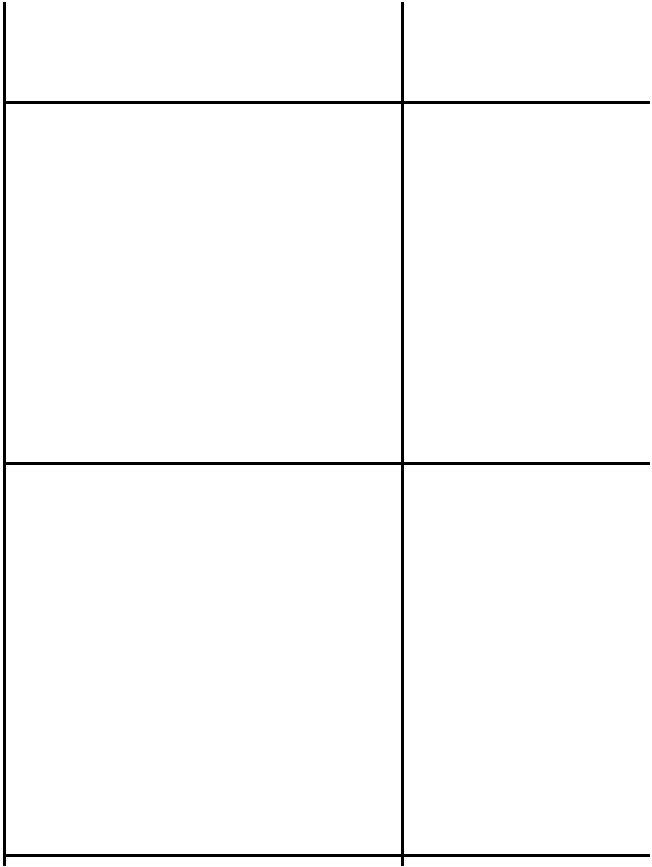


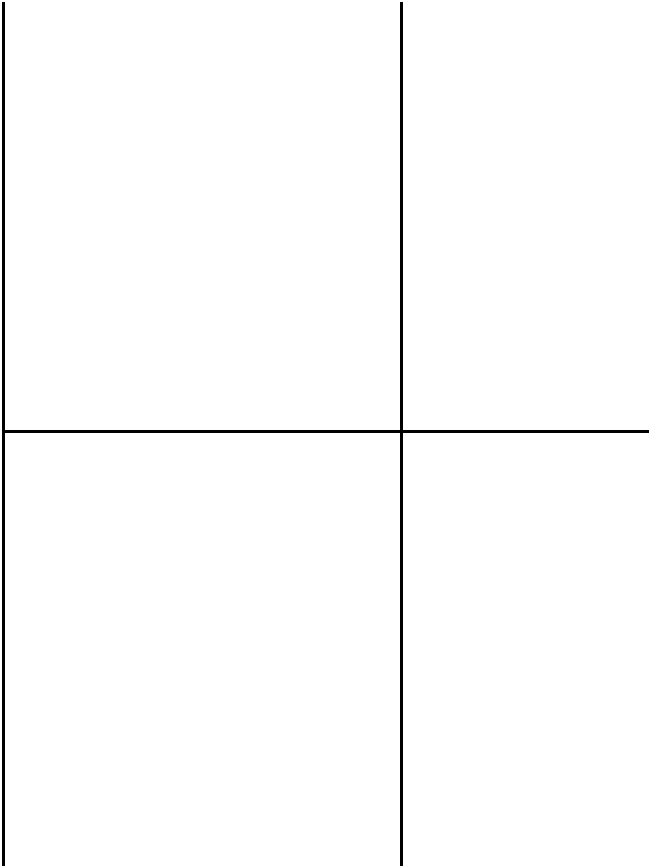














Exchanging Identity Data











Principles for Identity Data

Conclusion

Chapter 17. Inte Frameworks for Identity

Principles of a Good IF

Contents of an Identity IF

XML

TCP/IP

SAML

WS-Security

Encryption Algorithms

LDAP

DNS (RFC 1035)

IPSEC (RFC 2402/4)

SSL v3/TLS

External Standards

Identity Interoperability Framework

Software Standards

Hardware Standards

Figure 17-1. An interoperability framework provides the foundation for the policy stack

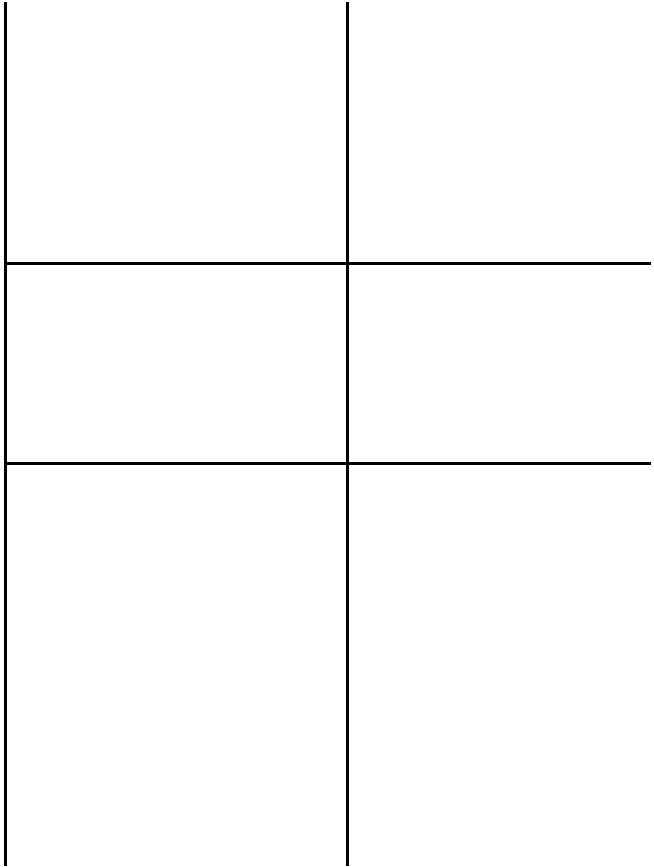
Standard Status

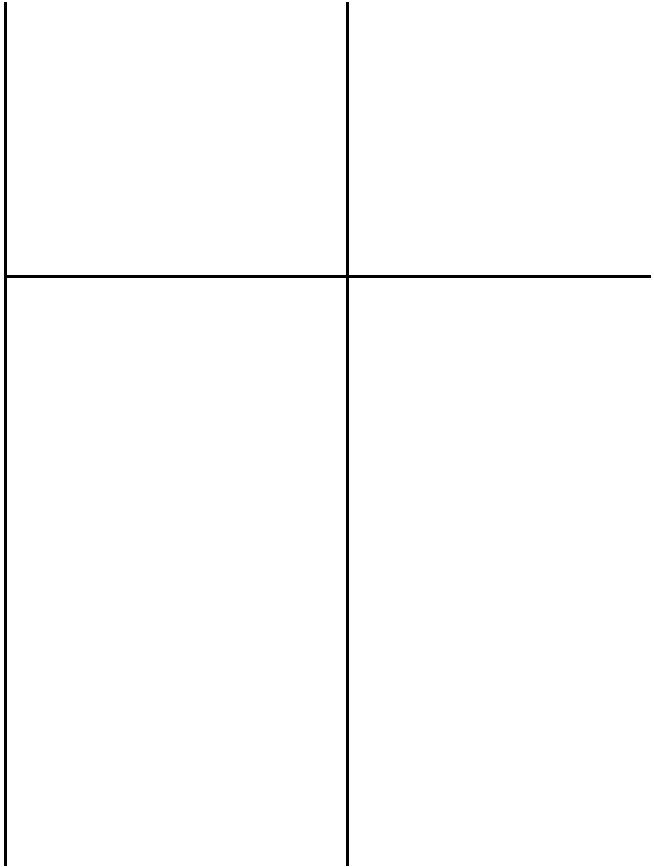
Listing Standards

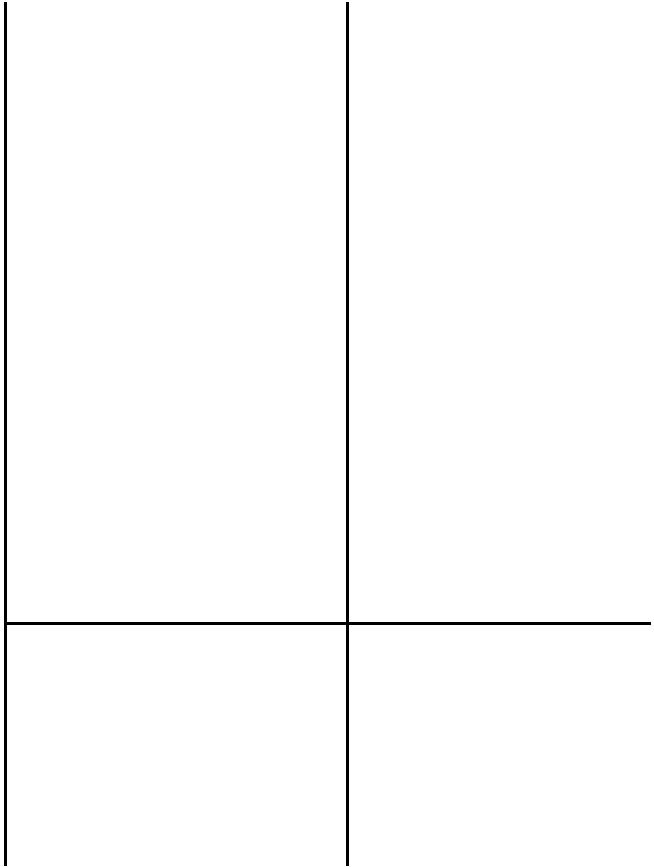
Example Interoperability Framework

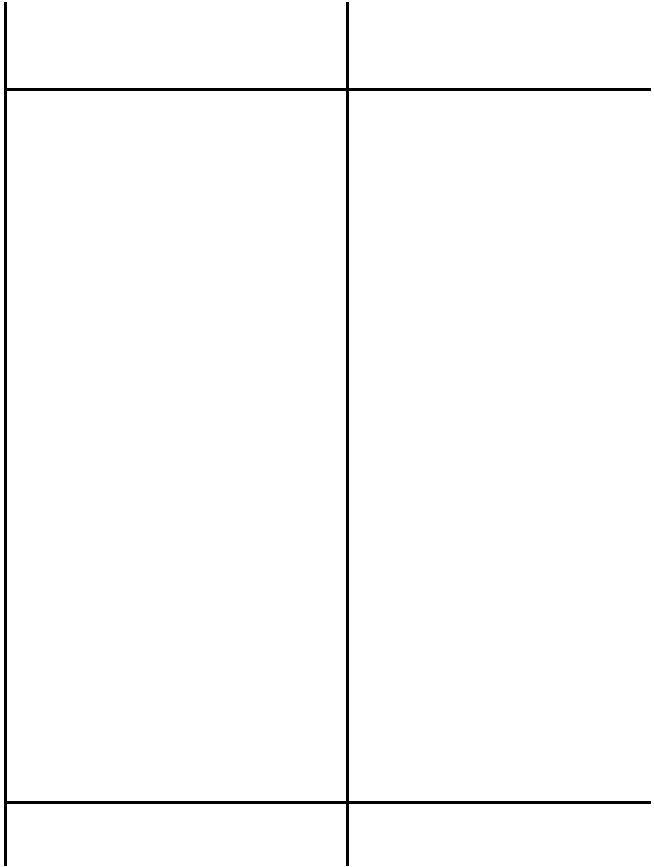
Table 17-1. Portion of an interoperability framework

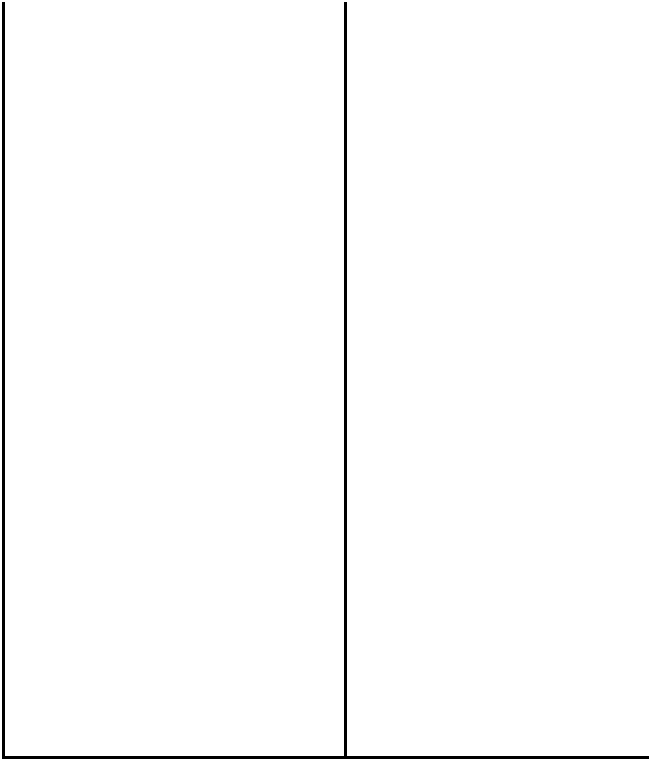
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A Word of Warning

Conclusion

Chapter 18. Identifier Policies

The Policy Stack

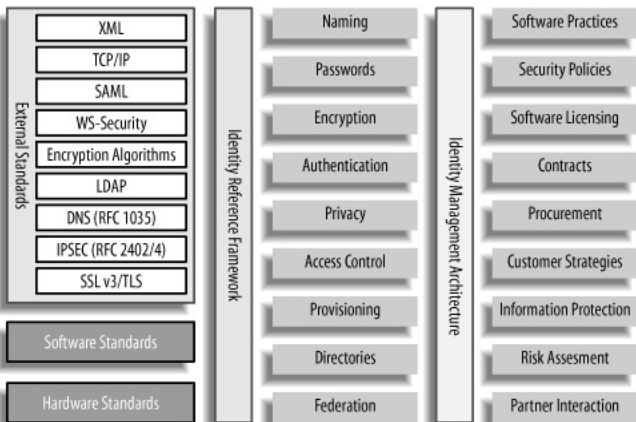


Figure 18-1. The policy stack with the interoperability framework as its foundation

Attributes of a Good Identity Policy

Determining Policy Needs



Business Inspired Projects and

Processes





Security Considerations

Meeting External Requirements

Feedback on Existing Policies

Writing Identity Policies















○

○

○







Policy Outline

An Identity Policy Suite

Naming and Certificates

Passwords





Encryption and Digital Signatures

Directories

Privacy

Authentication

Access Control





Provisioning

Federation

The Policy Review Framework







Assessing Identity Policies

Enforcement











Procedures

Conclusion

Chapter 19. Identifier Management Reference Architectures

Reference Architectures

Benefits and Pitfalls









Reference Architecture Best Practices

Using a Reference Architecture

Components of a Reference Architecture







Technical Position Statements

Making Decisions About Technical Positions

Consolidated Infrastructure Blueprint

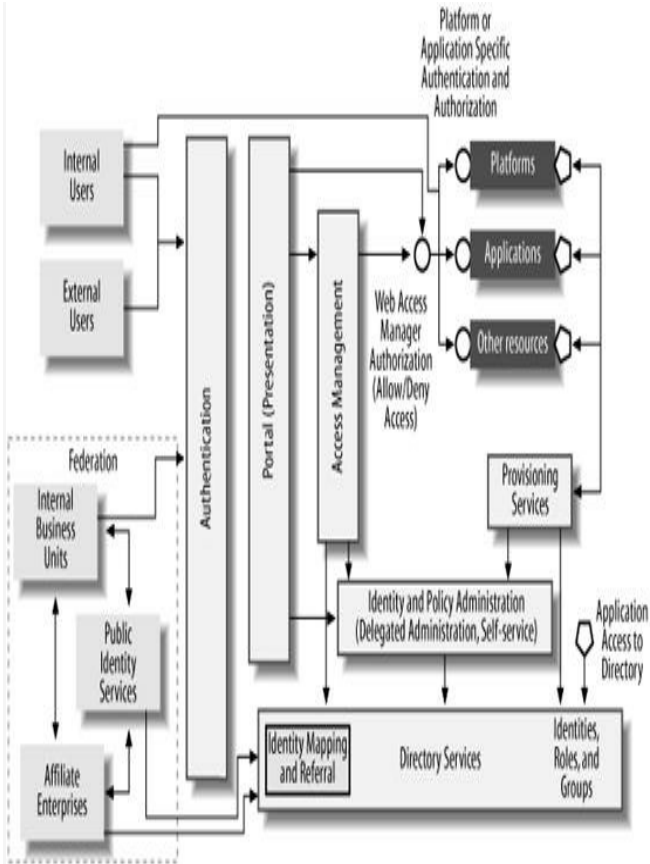


Figure 19-1. A consolidated infrastructure blueprint

Goal State CIBs

System Reference Architectures

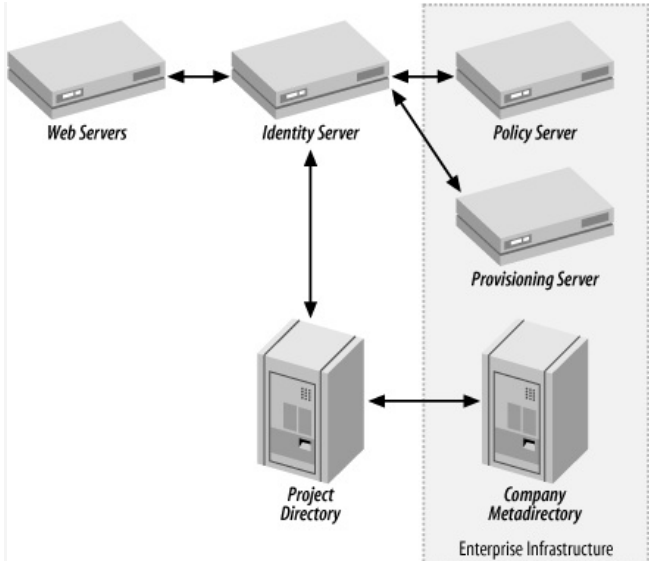


Figure 19-2. System reference architecture for web server authentication and authorization

Conclusion

Chapter 20. Building an Identity Management Architecture

Scoping the Process

Which Projects Are Enterprise Projects?

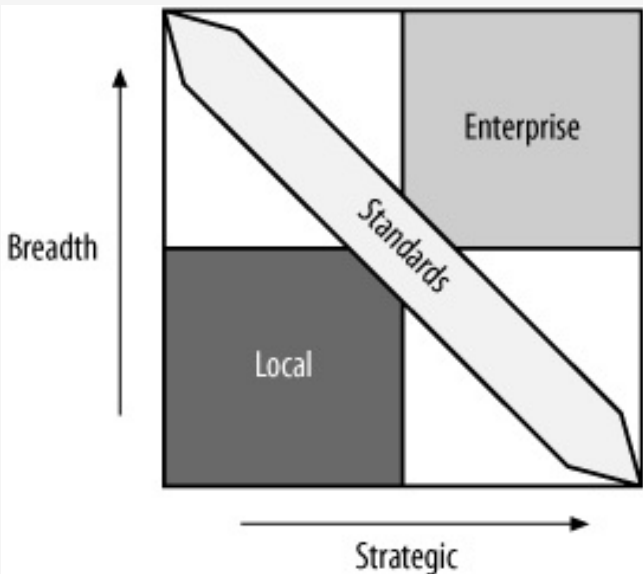


Figure 20-1. Enterprise decision matrix







Sequencing the IMA Effort

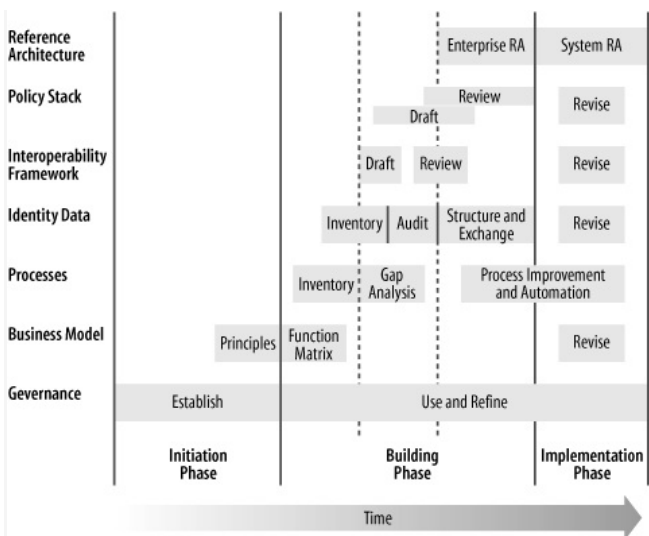


Figure 20-2. Phases in building an IMA

A Piece at a Time

Conclusion: Dispelling IMA Myths

Colophon

